

Kane-Mele Hubbard model and SG135 write-up

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1 Aims

Let's first review the general procedure for diagonalizing a quadratic Hamiltonian and then in the next three sections apply it to three examples - an s-wave superconductor ([1, p.266-271]), the Kane-Mele Hubbard Model from [2], and the Double-Dirac Semi-Metal in [3].

2 General Procedure

For this section I'll follow [4]. Consider a general Hamiltonian quadratic in either fermion or boson creation/annihilation operators:

$$H = \sum_{i,j} \xi_{ij} c_i^\dagger c_j + \frac{1}{2} \Delta_{ij} c_i^\dagger c_j^\dagger + \frac{1}{2} \Delta_{ij}^* c_j c_i \quad (2.1)$$

To solve this system, we can follow these steps:

1. Write down a Nambu spinor.
2. Write down the Hamiltonian Kernel.
3. Diagonalize the Hamiltonian Kernel (or a related Kernel for bosons).
4. Normal order to obtain the quasi-particle Hamiltonian.

2.1 Writing down the Nambu spinor

A conventional choice for the Nambu spinor is:

$$\psi = \begin{pmatrix} \mathbf{c} \\ \mathbf{c}^\dagger \end{pmatrix} \quad (2.2)$$

Where the vector $\mathbf{c}$ has components running over the degrees of freedom of the system. Since the Nambu-spinor has twice the degrees of freedom of the system, this representation doubles the Fock space of the system. We'll see that this means that only half of the eigenspectrum corresponds to the excitation spectrum of the system.

For a system which has non-spin degrees of freedom (i.e sublattice, valley, orbital etc...) $\propto$ a

conventional choice of Nambu spinor orders the non-spin degrees of freedom first¹:

$$\psi = \begin{pmatrix} c_{\alpha\sigma} \\ c_{\alpha(-\sigma)} \\ c_{\alpha\sigma}^\dagger \\ c_{\alpha(-\sigma)}^\dagger \end{pmatrix} \quad (2.3)$$

2.2 Writing down the Hamiltonian kernel

The Hamiltonian kernel $\mathcal{H}$ is the matrix that appears when we express the Hamiltonian as:

$$H = \psi^\dagger \mathcal{H} \psi + \text{c-number term} \quad (2.4)$$

To express our starting Hamiltonian eq. (2.1) in this form, we need the Hamiltonian kernel to have a particular structure. Let A, B, C, D be $N \times N$ block matrices. Then: In general, to double the Fock space we need to give the Nambu spinor a structure:

$$\psi = \begin{pmatrix} \mathbf{c} \\ \mathbf{c}^\dagger \end{pmatrix} \quad (2.5)$$

This means that we can express the Hamiltonian H in terms of a Hamiltonian kernel $\mathcal{H}$ and a constant term:

$$\sum_{ij} \psi^\dagger \mathcal{H} \psi = \begin{pmatrix} \mathbf{c}^\dagger & \mathbf{c} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \mathbf{c} \\ \mathbf{c}^\dagger \end{pmatrix} = \sum_{ij} c_i^\dagger A_{ij} c_j + c_i D_{ij} c_j^\dagger + c_i^\dagger B_{ij} c_j^\dagger + c_i C_{ij} c_j \quad (2.6)$$

In order for this term to be Hermitian we need:

$$(A_{ij} c_i^\dagger c_j)^\dagger = A_{ij}^* c_j^\dagger c_i = \zeta A_{ij}^* c_i c_j^\dagger + A_{ij}^* [c_i, c_j^\dagger] \stackrel{!}{=} D_{ij} c_i c_j^\dagger \iff \zeta A^* + \zeta \text{Tr } A^* = D \quad (2.7)$$

$$(B_{ij} c_i^\dagger c_j^\dagger)^\dagger = B_{ij}^* c_j c_i = \zeta B_{ij}^* c_i c_j \stackrel{!}{=} C_{ij} c_i c_j \iff \zeta B^* = C \quad (2.8)$$

¹Notice that Altland and Simons use a different convention on p. 271! They use $\psi_{\mathbf{k}} = (c_{\mathbf{k}\uparrow}, c_{-\mathbf{k}\downarrow}^\dagger)$. This is equivalent to just a basis re-ordering for the basis in the convention I'm using, but it makes it more difficult to identify the para-unitary transformation you need to diagonalize bosonic operators whilst preserving the canonical commutation relations. Notice also that the convention here gives you two copies of every term in the Hamiltonian whereas the AS convention only gives you once. You can always choose your basis that way which eliminates the doubling of the Fock space.

Where $\zeta = +1$ for bosons and $\zeta = -1$ for fermions². If we make these definitions, then:

$$\sum_{ij} \psi^\dagger \mathcal{H} \psi = \sum_{ij} 2A_{ij} c_i^\dagger c_j + B_{ij} c_i^\dagger c_j^\dagger + h.c. + \zeta \text{Tr}\{A^*\} \quad (2.9)$$

Then the Hamiltonian kernel:

$$\begin{pmatrix} \xi & \Delta \\ \zeta \Delta^* & \zeta \xi^* \end{pmatrix} \quad (2.10)$$

gives:

$$\sum_{ij} \psi^\dagger \mathcal{H} \psi = \sum_{ij} 2(\xi_{ij} c_i^\dagger c_j + \frac{1}{2} \Delta_{ij} c_i^\dagger c_j^\dagger) + h.c. + \zeta \text{Tr}\{\xi^*\} \quad (2.11)$$

And so our original Hamiltonian eq. (2.1):

$$H = \sum_{i,j} \xi_{ij} c_i^\dagger c_j + \frac{1}{2} \Delta_{ij} c_i^\dagger c_j^\dagger + \frac{1}{2} \Delta_{ij}^* c_j c_i \quad (2.12)$$

can be expressed as:

$$H = \sum_{ij} \frac{1}{2} \psi_i^\dagger \mathcal{H}_{ij} \psi_j - \zeta \text{Tr}\{\xi^*\} = \sum_{ij} \frac{1}{2} \psi_i^\dagger \mathcal{H}_{ij} \psi_j - \frac{1}{2} \zeta \text{Tr}\{\xi\} \quad (2.13)$$

Since ξ is Hermitian.

2.3 Diagonalizing the Hamiltonian kernel

To diagonalize the Hamiltonian, we need to:

1. Find quasi-particle operators consisting of a linear combination of creation and annihilation operators in which the Hamiltonian is diagonal,
2. Ensure that the quasi-particle operators satisfy canonical commutation relations.

For the fermionic case, this is equivalent to diagonalizing $\mathcal{H}$ by a unitary transformation. The bosonic case is more subtle and was worked out in [5] and [6]. The result is that we need to diagonalize a Hamiltonian $\lambda^z \mathcal{H}$ with λ^z the usual z Pauli-matrix in the basis of block diagonal matrices, i.e.

$$\lambda^z = \begin{pmatrix} \mathbb{1}_N & 0 \\ 0 & -\mathbb{1}_N \end{pmatrix} \quad (2.14)$$

²Note that since you have to consider each $c_i c_j$ term separately, when we take the Hermitian conjugate $(B_{ij} c_i^\dagger c_j^\dagger)^\dagger$ the B matrix indices don't transpose since we're considering just that term.

Assuming the existence of a (Bogoliubov) transformation η which acts on the Nambu spinor ψ to give a spinor of quasi-particle operators $\tilde{\psi}$:

$$\tilde{\psi} \equiv \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix} = \eta \begin{pmatrix} \mathbf{c} \\ \mathbf{c}^\dagger \end{pmatrix} \quad (2.15)$$

in terms of which the Hamiltonian kernel $\mathcal{H}$ ($\lambda^z \mathcal{H}$ for bosons) is diagonal³:

$$\psi^\dagger \mathcal{H} \psi = \tilde{\psi}^\dagger (\eta^{-1})^\dagger \mathcal{H} \eta^{-1} \tilde{\psi} = \sum_n \epsilon_n b_n^\dagger b_n + \epsilon_{-n} b_{-n} b_{-n}^\dagger = \sum_n \epsilon_n (b_n^\dagger b_n - b_{-n} b_{-n}^\dagger) \quad (2.16)$$

2.4 Normal ordering quasi-particle operators

We can then use the relation⁴ $b_n = -b_{-n}^\dagger$ and the commutation relation for the b_{-n} 's - $[b_{-m}, b_{-n}^\dagger] = -\delta_{mn}$ (anti-commutator for fermions) to get :

$$\psi^\dagger \mathcal{H} \psi = \sum_n 2\epsilon_n b_n^\dagger b_n - \frac{1}{2} \sum_n \epsilon_n \quad (2.17)$$

Substituting this into the expression for the Hamiltonian we get:

$$H = \sum_{ij} \frac{1}{2} \psi_i^\dagger \mathcal{H}_{ij} \psi_j - \frac{1}{2} \zeta \text{Tr}\{\xi\} = \sum_n \epsilon_n b_n^\dagger b_n - \frac{1}{2} \left(\sum_n \epsilon_n + \zeta \text{Tr}\{\xi\} \right) \quad (2.18)$$

This last step can be subtle since it might not be obvious after we've diagonalized which are the positive and which the negative energy eigenvalues.

2.5 Summary

So to diagonalize the Hamiltonian, we:

1. Diagonalize the Hamiltonian kernel $\mathcal{H}$ for fermions.
2. Diagonalize $\lambda^z \mathcal{H}$ for bosons.

For both we should get:

1. A complex number (i.e no operators) contribution to the ground state energy $-\sum_n \epsilon_n$ from the (anti-)commutator.

³I've just assumed here that the spectrum of $\mathcal{H}$ partitions into a part with a positive energy ϵ_n created by $b_n^\dagger b_n$ and a part with the negative energy $\epsilon_{-n} = -\epsilon_n$ created by $b_{-n}^\dagger b_{-n}$. This requires some work, which is covered in [4], [6] and [7].

⁴Again, I'm just assuming it following the references.

2. The spectrum should have both positive and negative branches ϵ_n and $\epsilon_{-n} = -\epsilon_n$. Equation eq. (2.17) tells us that only the positive energies correspond to excitation energies of the quasiparticles. Identifying which these are will allow us to identify a BCS/BEC cross-over for the bosonic case and help us to reproduce [2].

We'll now illustrate this procedure on three examples.

3 Warm-up: Mean-field description of superconductor

When the attraction between two electrons becomes strong enough that they form a Cooper pair, a Fermi liquid is no longer a valid ground state of a metal. Instead, we have a macroscopic number of Cooper pairs. If we expand around this ground state⁵, we get the BdG Hamiltonian [1, p. 266-271]:

$$H - \mu N = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \left((\sigma)^\zeta \Delta(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-(\sigma)}^\dagger + h.c. \right) \quad (3.1)$$

Where we have defined $\xi_{\mathbf{k}}$ for convenience as the part of the Hamiltonian diagonal in the spin degrees of freedom σ (i.e the kinetic energy the electrons would have in the absence of pairing.)

$$\xi_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} - \mu \quad (3.2)$$

Notice that the $(\sigma)^\zeta$ comes from the fact that the Fourier transform of the pairing terms:

$$\sum_{\langle i,j \rangle} \Delta c_{i\uparrow}^\dagger c_{j\downarrow}^\dagger + \Delta c_{j\downarrow}^\dagger c_{i\uparrow}^\dagger \rightarrow \sum_{\mathbf{k}} \Delta g(\mathbf{k}) c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + \Delta g(-\mathbf{k}) c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \quad (3.3)$$

$$= \sum_{\mathbf{k}} \Delta g(\mathbf{k}) c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + \Delta g(\mathbf{k}) c_{-\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}\downarrow}^\dagger \quad (3.4)$$

$$= \sum_{\mathbf{k}} \Delta g(\mathbf{k}) c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + \zeta \Delta g(\mathbf{k}) c_{\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\uparrow}^\dagger \quad (3.5)$$

$$= \sum_{\mathbf{k}\sigma} (\sigma)^\zeta \Delta g(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-\sigma}^\dagger \quad (3.6)$$

Let's now find the eigensystem of this Hamiltonian by following the procedure in the previous section:

1. Artificially double the Fock space by writing down a Nambu spinor,
2. Write the Hamiltonian in terms of a Hamiltonian kernel acting on the Nambu spinors,
3. Diagonalize the Hamiltonian kernel,
4. Normal order the quasi-particle operators by identifying the positive energy eigenvalues.

⁵Equivalently, perform a Hubbard-Stratonovich decoupling of the path integral - which is the same setup as for the Coleman slave-boson approach.

3.1 Writing down the Nambu spinor

Following the convention of the previous section for the Nambu spinor eq. (2.2) we can choose:

$$\psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}(\downarrow)} \\ c_{\mathbf{k}\uparrow}^\dagger \\ c_{-\mathbf{k}(\downarrow)}^\dagger \end{pmatrix} \quad (3.7)$$

3.2 Writing down the Hamiltonian kernel

3.2.1 Kinetic Term

To write down the Hamiltonian kernel we need to identify the block matrices ξ and Δ . We can start by considering:

$$\sum_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix} \begin{pmatrix} \xi_{\mathbf{k}\uparrow} & 0 \\ 0 & \xi_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow} \end{pmatrix} = \sum_{\mathbf{k}} \xi_{\mathbf{k}\uparrow} c_{\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow} c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\downarrow} \quad (3.8)$$

$$= \sum_{\mathbf{k}} \xi_{\mathbf{k}\uparrow} c_{\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}\uparrow} + \sum_{-\mathbf{k}} \xi_{\mathbf{k}\downarrow} c_{\mathbf{k}\downarrow}^\dagger c_{\mathbf{k}\downarrow} = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} \quad (3.9)$$

$$= D\sigma^0 + \Delta_0\sigma^z \quad (3.10)$$

Where we define:

$$D \equiv \frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2} \quad (3.11)$$

$$\Delta_0 \equiv \frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2} \quad (3.12)$$

3.2.2 Pairing Term

Δ is a little trickier. Consider

$$\Delta = \begin{pmatrix} 0 & -\Delta \\ -\Delta & 0 \end{pmatrix} \quad (3.13)$$

Then for the top-right block we get:

$$\sum_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix} \begin{pmatrix} 0 & -\Delta \\ -\Delta & 0 \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix} = \sum_{\mathbf{k}} - \left(\underbrace{\Delta c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger}_{\text{Correct}} + \Delta \underbrace{c_{-\mathbf{k}\downarrow}^\dagger c_{\mathbf{k}\uparrow}^\dagger}_{\text{Wrong sign in } \mathbf{k}} \right) \quad (3.14)$$

If we compare the second term to the term in the original Hamiltonian eq. (3.1):

$$\text{Our term} = \sum_{\mathbf{k}} -\Delta c_{-\mathbf{k}\downarrow}^\dagger c_{\mathbf{k}\uparrow}^\dagger \quad (3.15)$$

$$\text{Term we want} = \sum_{\mathbf{k}} +\Delta c_{\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\uparrow}^\dagger \quad (3.16)$$

But we can re-index the sum $\mathbf{k} \rightarrow -\mathbf{k}$ (remembering that Δ is, in general, a function of $\mathbf{k}$):

$$\sum_{\mathbf{k}} -\Delta(\mathbf{k}) c_{-\mathbf{k}\downarrow}^\dagger c_{\mathbf{k}\uparrow}^\dagger = \sum_{-\mathbf{k}} -\Delta(-\mathbf{k}) c_{\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\uparrow}^\dagger = \sum_{\mathbf{k}} -\Delta(-\mathbf{k}) c_{\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\uparrow}^\dagger \quad (3.17)$$

Where the last equality follows because the sum over $\mathbf{k}$ ranges over the same values as the sum over $-\mathbf{k}$. So to get the right expression we need the off-diagonal block to be:

$$\Delta = \begin{pmatrix} 0 & -\Delta(\mathbf{k}) \\ -\zeta\Delta(-\mathbf{k}) & 0 \end{pmatrix} \quad (3.18)$$

If the pairing is at least time-reversal invariant we have⁶ $\Delta(\mathbf{k}) = \Delta^*(-\mathbf{k})$. If the pairing is inversion symmetric then we should have $\Delta(\mathbf{k}) = \Delta(-\mathbf{k})$. The first case arises in the Kane-Mele Hubbard model [2], and the second when taking the slave boson theory for a tight binding model with space group SG135 [3].

Let's check that the bottom-left block in the Hamiltonian kernel $\zeta\mathcal{H}$ gives us what we expect:

$$\sum_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k}\uparrow} & c_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{pmatrix} 0 & -\zeta\Delta^*(\mathbf{k}) \\ -\Delta^*(-\mathbf{k}) & 0 \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow} \end{pmatrix} = \sum_{\mathbf{k}} -(\zeta\Delta^*(\mathbf{k})c_{\mathbf{k}\uparrow}c_{-\mathbf{k}\downarrow} + \Delta^*(-\mathbf{k})c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow}) \quad (3.19)$$

Comparing to the term in the Hamiltonian:

$$\text{Our term} = \sum_{\mathbf{k}} -(\zeta\Delta^*(\mathbf{k})c_{\mathbf{k}\uparrow}c_{-\mathbf{k}\downarrow} + \Delta^*(-\mathbf{k})c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow}) \quad (3.20)$$

$$\text{Term we want} = \sum_{\mathbf{k}} -(\Delta^*(\mathbf{k})c_{-\mathbf{k}\downarrow}c_{\mathbf{k}\uparrow} - \zeta\Delta^*(\mathbf{k})c_{-\mathbf{k}\uparrow}c_{\mathbf{k}\downarrow}) \quad (3.21)$$

⁶I think this is generally true for pairing that is a scalar - but might not be true for a spin dependent pairing - Should double check

$\zeta = 1$ for bosons and $\zeta = -1$ for fermions, so we can commute the operators through each other at the price of the ζ . When we do that we see that the terms are the same as we'd expect:

$$\sum_{\mathbf{k}} -(\zeta \Delta^*(\mathbf{k}) c_{\mathbf{k}\uparrow} c_{-\mathbf{k}\downarrow} - \Delta^*(-\mathbf{k}) c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow}) = -\sum_{\mathbf{k}} \Delta^*(\mathbf{k}) c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} + \zeta \sum_{\mathbf{k}} \Delta^*(-\mathbf{k}) c_{\mathbf{k}\uparrow} c_{-\mathbf{k}\downarrow} \quad (3.22)$$

$$= -\sum_{\mathbf{k}} \Delta^*(\mathbf{k}) c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} + \zeta \sum_{-\mathbf{k}} \Delta^*(\mathbf{k}) c_{-\mathbf{k}\uparrow} c_{\mathbf{k}\downarrow} = -\sum_{\mathbf{k}\sigma} (\sigma)^\zeta c_{-\mathbf{k}-\sigma} c_{\mathbf{k}\sigma} \quad (3.23)$$

$$= -\left(\sum_{\mathbf{k}\sigma} (\sigma)^\zeta c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-\sigma}^\dagger \right)^\dagger \quad (3.24)$$

And so we have verified that:

$$\Delta = \begin{pmatrix} 0 & -\Delta(\mathbf{k}) \\ -\zeta \Delta(-\mathbf{k}) & 0 \end{pmatrix} \quad (3.25)$$

is the right choice for this Hamiltonian.

We can decompose this into Pauli matrices in a similar way to what we did for the kinetic term eq. (3.8):

$$\begin{pmatrix} 0 & -\Delta(\mathbf{k}) \\ -\zeta \Delta(-\mathbf{k}) & 0 \end{pmatrix} \equiv \begin{cases} -\Delta_I \sigma^x - \Delta_{II} (i\sigma^y), & \text{for } \zeta = 0 \text{ (bosons)} \\ -\Delta_I (i\sigma^y) - \Delta_{II} \sigma^x & \text{for } \zeta = 1 \text{ (fermions)} \end{cases} \quad (3.26)$$

Where:

$$\Delta_I \equiv \frac{\Delta(\mathbf{k}) + \Delta(-\mathbf{k})}{2} \quad (3.27)$$

$$\Delta_{II} \equiv \frac{\Delta(\mathbf{k}) - \Delta(-\mathbf{k})}{2} \quad (3.28)$$

Note that for the case in which $\Delta(\mathbf{k}) = \Delta(-\mathbf{k})$ (the case of the tight binding model for SG135 in [3] with no spin orbit coupling) this simplifies to $\Delta_I = \Delta$, $\Delta_{II} = 0$ and for the case in which $\Delta(\mathbf{k}) = \Delta^*(-\mathbf{k})$ we have $\Delta_I = \text{Re}[\Delta]$ and $\Delta_{II} = \text{Im}[\Delta]$.

In the case of an s-wave superconductor, $\Delta(\mathbf{k})$ is just a (complex number) constant, Δ , and hence the pairing term in the Hamiltonian kernel is:

$$\Delta = \begin{cases} -\Delta \sigma^x, & \text{for } \zeta = 0 \text{ (bosons)} \\ -\Delta i\sigma^y, & \text{for } \zeta = 1 \text{ (fermions)} \end{cases} \quad (3.29)$$

3.2.3 Hamiltonian Kernel

Putting these results together, the Hamiltonian Kernel $\mathcal{H}_f$ for the *fermionic* Hamiltonian :

$$H - \mu N = \sum_{\mathbf{k}} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \left(\Delta^* c_{-\mathbf{k}(-\sigma)} c_{\mathbf{k}(\sigma)} + \Delta c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}(-\sigma)}^\dagger \right) \quad (3.30)$$

is (Using the general structure eq. (2.10)):

$$\mathcal{H}_f = \left[\begin{array}{cc|cc} D + \Delta_0 & 0 & 0 & -\Delta \\ 0 & D - \Delta_0 & \Delta & 0 \\ \hline 0 & \Delta^* & -(D + \Delta_0) & 0 \\ -\Delta^* & 0 & 0 & -(D - \Delta_0) \end{array} \right] \quad (3.31)$$

Or, in terms of Pauli matrices:

$$\mathcal{H}_f = D\lambda^z\sigma^0 + \Delta_0\lambda^z\sigma^z - (Re[\Delta]i\lambda^y i\sigma^y + iIm[\Delta]\lambda^x i\sigma^y) \quad (3.32)$$

$$= D\lambda^z\sigma^0 + \Delta_0\lambda^z\sigma^z + (Re[\Delta]\lambda^y\sigma^y + Im[\Delta]\lambda^x\sigma^y) \quad (3.33)$$

Notice the λ^x in the imaginary part of Δ since the complex conjugation cancels out with the ζ from the commutator.

For bosons, the Hamiltonian kernel $\mathcal{H}_b$ is given by:

$$\mathcal{H}_b = \left[\begin{array}{cc|cc} D + \Delta_0 & 0 & 0 & -\Delta \\ 0 & D - \Delta_0 & -\Delta & 0 \\ \hline 0 & -\Delta^* & (D + \Delta_0) & 0 \\ -\Delta^* & 0 & 0 & (D - \Delta_0) \end{array} \right] \quad (3.34)$$

Or, in Pauli matrices:

$$\mathcal{H}_b = D\lambda^0\sigma^0 + \Delta_0\lambda^0\sigma^z - (Re[\Delta]\lambda^x\sigma^x + iIm[\Delta]i\lambda^y\sigma^x) \quad (3.35)$$

$$= D\lambda^0\sigma^0 + \Delta_0\lambda^0\sigma^z - (Re[\Delta]\lambda^x\sigma^x - Im[\Delta]\lambda^y\sigma^x) \quad (3.36)$$

And we can verify that:

$$\frac{1}{2}\psi_{\mathbf{k}}^\dagger \mathcal{H}_b \psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow}^\dagger & c_{\mathbf{k}\uparrow} & c_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{pmatrix} D + \Delta_0 & 0 & 0 & -\Delta \\ 0 & D - \Delta_0 & -\Delta & 0 \\ 0 & -\Delta^* & D + \Delta_0 & 0 \\ -\Delta^* & 0 & 0 & D - \Delta_0 \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow} \\ c_{\mathbf{k}\uparrow}^\dagger \\ c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix} \quad (3.37)$$

$$= \sum_{\sigma} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \left(\Delta c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-\sigma}^\dagger + \Delta^* c_{\mathbf{k}\sigma} c_{-\mathbf{k}-\sigma} \right) \quad (3.38)$$

since $\xi_{\mathbf{k}\uparrow} = D + \sigma_0$, $\xi_{\mathbf{k}\downarrow} = D - \sigma_0$.

3.3 Diagonalizing the Hamiltonian

The easiest way to diagonalize is to follow an algorithm:

1. Identify any set of Pauli matrices which anti-commute with all other elements in the set, but commute with all elements outside the set⁷.
2. Remove any such sets and diagonalize the set individually. Since the set commutes with the rest of the terms, the eigenvalue will be the sum of the eigenvalues of this set and the others.
3. Square the remaining terms, and then repeat steps 1 and 2 until the remaining terms are proportional to the identity.

If this process terminates, then you can use it to find the eigenvalues⁸. Notice also that this gives us the structure we expect from the general procedure in section 2. This follows since when squaring the Hamiltonian we get a $\pm$ on taking the square root of the spectrum, which in turns gives us the positive and negative branches.

Conveniently, there is an easy way to check for step 1. If we consider each term as a node in a network, and draw vertices between any two terms that anti-commute, the maximal sets that commute with all others are just any ‘islands’ disconnected from the rest of the network. We will see this in the examples.

⁷Notice that this is stronger than what you need - I think you can have a set of matrices where the overall term commutes with all the others, but not each individual term.

⁸It would be good to know general conditions under which it work, but I don’t know them

3.3.1 Fermionic case

For the fermionic case we need to diagonalize:

$$\mathcal{H}_f = D\lambda^z\sigma^0 + \Delta_0\lambda^z\sigma^z - (Re[\Delta]\lambda^y\sigma^y + Im[\Delta]\lambda^x\sigma^y) \quad (3.39)$$

Let's hope the diagonalization procedure works in this case. We start by calculating all anti-commutators in $\mathcal{H}_f$ and drawing the network - fig. 1 The network conveniently splits into two

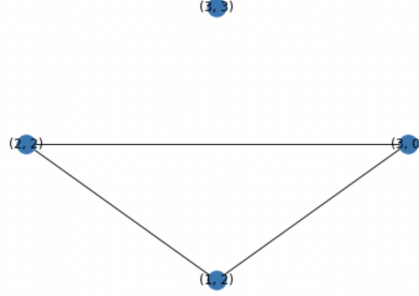


Figure 1: The network of anti-commutators of $\mathcal{H}$. Each node is a tensor product of Pauli matrices and an edge exists between any two nodes which anti-commute.

“islands”. This allows us to remove the spin splitting term $\Delta_0\lambda^z\sigma^z$. It diagonalizes trivially to give $\pm\Delta_0$.

We then square the remaining kernel:

$$\mathcal{H}'_f = D\lambda^z\sigma^0 - (Re[\Delta]\lambda^y\sigma^y + Im[\Delta]\lambda^x\sigma^y) \quad (3.40)$$

Looking at the network, we know already that the remaining matrices all anti-commute, so the square of the remaining kernel $\mathcal{H}'_f$ will end up being proportional to the identity - i.e:

$$\mathcal{H}_f'^2 = (D^2 + Re[\Delta]^2 + Im[\Delta]^2)\lambda^0\sigma^0 \quad (3.41)$$

And so the eigenvalues of $\mathcal{H}'_f$ are:

$$E'^2 = \pm\sqrt{D^2 + Re[\Delta]^2 + Im[\Delta]^2} = \pm\sqrt{D^2 + |\Delta|^2} \quad (3.42)$$

Since we know $\mathcal{H}'_f$ commutes with the $\Delta_0\lambda^z\sigma^z$ the resulting spectrum of $\mathcal{H}_f$ is just the sum of the eigenvalues:

$$E = \pm\Delta_0 \pm \sqrt{D^2 + |\Delta|^2} \quad (3.43)$$

This has exactly the structure we expect - a positive and a negative branch. The positive branch corresponds to the excitation energies of the quasi-particles. Replacing D, Δ_0 and Δ with their

definitions in terms of the parameters of the Hamiltonian the spectrum is:

$$E = \pm_1 \left(\frac{\xi_{k\uparrow} - \xi_{k\downarrow}}{2} \right) \pm_2 \sqrt{\frac{1}{4}(\xi_{k\uparrow} + \xi_{k\downarrow})^2 + |\Delta|^2} \quad (3.44)$$

Which reduces to the result in Altland-Simons [1, p. 272]:

$$\sqrt{\xi_{\mathbf{k}}^2 + \Delta^2} \quad (3.45)$$

When there is no spin-splitting and the kinetic terms is inversion symmetric: $\xi_{\mathbf{k}\uparrow} = \xi_{-\mathbf{k}\downarrow}$.

3.3.2 Normal ordering the quasi-particle Hamiltonian

Recall that in the final step of the general procedure section 2 we wrote down that the Hamiltonian is given by:

$$H = \sum_{ij} \frac{1}{2} \psi_i^\dagger \mathcal{H}_{ij} \psi_j - \zeta \text{Tr}\{\xi\} = \sum_n \epsilon_n b_n^\dagger b_n - \frac{1}{2} \left(\sum_n \epsilon_n + \zeta \text{Tr}\{\xi\} \right) \quad (3.46)$$

Since all of the quasi-particle excitation energies ϵ_n are positive by definition, the ground state has an energy:

$$-\frac{1}{2} \zeta \text{Tr}\{\xi\} - \frac{1}{2} \sum_n \epsilon_n \quad (3.47)$$

So in the fermionic case, our ground state energy is given by:

$$\frac{1}{2} \text{Tr}\{\xi\} - \frac{1}{2} \sum_n \epsilon_n = \frac{1}{2} \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} \sqrt{\frac{1}{4}(\xi_{\mathbf{k}\uparrow} + \xi_{\mathbf{k}\downarrow})^2 + |\Delta|^2} \quad (3.48)$$

Which again is the generalization of the result in Altland and Simons.

3.3.3 Summary for fermions

To summarize, we have found that the **fermionic** Hamiltonian:

$$H_f = H - \mu N = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \left(\sigma \Delta(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-(\sigma)}^\dagger + h.c. \right) \quad (3.49)$$

Can be canonically transformed into the form:

$$H_f = \sum_{\mathbf{k}\sigma} \epsilon_{(\sigma\mathbf{k})\sigma} \alpha_{\mathbf{k}\sigma}^\dagger \alpha_{\mathbf{k}\sigma} + \frac{1}{2} \left(\sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} - \sum_{\mathbf{k}n} \epsilon_{\mathbf{k}n} \right) \quad (3.50)$$

With quasi-particle excitation energies ϵ_{kn} :

$$\epsilon_{kn} = \pm \frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2} + \sqrt{\left(\frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2}\right)^2 + |\Delta|^2} \quad (3.51)$$

And a ground state energy:

$$\frac{1}{2} \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} \sqrt{\frac{1}{4}(\xi_{\mathbf{k}\uparrow} + \xi_{\mathbf{k}\downarrow})^2 + |\Delta|^2} \quad (3.52)$$

The only thing left to do would be to find the eigenvalues corresponding to these energies, but I'll leave that for now.

Clearly, we didn't need all of this machinery to solve this problem but all of it will transfer over readily to the more complicated cases we will consider in the next two sections. We can use the properties of the results we've got here as a sanity check on those results.

3.3.4 Diagonalizing the bosonic case

We now do the same analysing for the bosonic case. The Hamiltonian kernel we found in eq. (3.32) $\mathcal{H}_b$:

$$\mathcal{H}_b = D\lambda_0\sigma^0 + \Delta_0\lambda^0\sigma^0 - (Re[\Delta]\lambda^x\sigma^x - Im[\Delta]\lambda^y\sigma^x) \quad (3.53)$$

In the general procedure, we quoted the result that to diagonalize the Bosonic Hamiltonian whilst preserving canonical commutation relations, we need to diagonalize $\lambda^z\mathcal{H}_b$. We have:

$$\lambda^z\mathcal{H}_b = D\lambda^z\sigma^z + \Delta_0\lambda^z\sigma^z - (Re[\Delta]i\lambda^y\sigma^x + Im[\Delta]i\lambda^x\sigma^x) \quad (3.54)$$

Notice that introduced a factor of i into the pairing. This will switch the sign of the pairing term in the spectrum, which is what we would expect. We can now proceed in exactly the same way as for the fermionic case. Plotting the network of anti-commutators in fig. 2 we see we have exactly the same structure as before (but note that the coefficients are different in Δ by a factor of i !). And so we can again read off the spectrum by squaring the kernel after removing the Δ_0 term which commutes with all the others. We then get for the spectrum:

$$E = \pm\Delta_0 \pm \sqrt{D^2 - |\Delta|^2} \quad (3.55)$$

As before, *only the positive energies* correspond to excitation energies. Identifying the positive energies is a little more subtle for the bosons. We have two cases:

$$E = \begin{cases} \sqrt{D^2 - |\Delta|^2} \pm \Delta_0, & \text{for } \sqrt{D^2 - |\Delta|^2} > |\Delta_0| \\ |\Delta_0| \pm \sqrt{D^2 - |\Delta|^2}, & \text{for } \sqrt{D^2 - |\Delta|^2} < |\Delta_0| \end{cases} \quad (3.56)$$

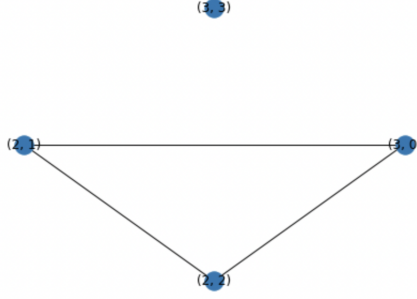


Figure 2: The network of anti-commutators of $\mathcal{H}$. Each node is a tensor product of Pauli matrices and an edge exists between any two nodes which anti-commute.

The first case occurs when the pairing is small and corresponds to the formation of a Bose-Einstein Condensate. The second case occurs when the pairing is large and correspond to the formation of Cooper pairs. This is the BEC/BCS crossover. We can qualitatively understand it by considering the system as having a choose of forming either Cooper pairs or condensing particles into the $k = 0$ state. Clearly Cooper pair formation will be favoured if the pairing is stronger, and so it corresponds to the second case. A helpful reference is [8].

3.3.5 Summary for bosons

To summarize, we have found that the **bosonic** Hamiltonian:

$$H_f = H - \mu N = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \left(\Delta(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{-\mathbf{k}-(\sigma)}^\dagger + h.c. \right) \quad (3.57)$$

Can be canonically transformed into the form:

$$H_f = \sum_{\mathbf{k}\sigma} \epsilon_{\mathbf{k}\sigma} \beta_{\mathbf{k}\sigma}^\dagger \beta_{\mathbf{k}\sigma} - \frac{1}{2} \left(\sum_{\mathbf{k}\sigma} \xi_{(\sigma\mathbf{k})\sigma} + \sum_{\mathbf{k}n} \epsilon_{\mathbf{k}n} \right) \quad (3.58)$$

With quasi-particle excitation energies $\epsilon_{\mathbf{k}n}$:

$$\epsilon_{\mathbf{k}n} = \pm \frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2} + \sqrt{\left(\frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2} \right)^2 - |\Delta|^2} \quad (3.59)$$

And a ground state energy:

$$-\frac{1}{2} \left(\sum_{\mathbf{k}\sigma} \xi_{(\sigma\mathbf{k})\sigma} + \sum_{\mathbf{k}} \sqrt{(\xi_{\mathbf{k}\uparrow} + \xi_{\mathbf{k}\downarrow})^2 - 4|\Delta|^2} \right) \quad (3.60)$$

For the case that $\sqrt{\left(\frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2}\right)^2 - |\Delta|^2} > \left|\frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2}\right|$. This corresponds to a ground state of Cooper pairs.

For the opposite case in which $\sqrt{\left(\frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2}\right)^2 - |\Delta|^2} < \left|\frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2}\right|$ we have a macroscopic BEC in the ground state and we get instead quasi-particle excitation energies:

$$\epsilon_{\mathbf{k}n} = \left| \frac{\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}}{2} \right| \pm \sqrt{\left(\frac{\xi_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}\downarrow}}{2}\right)^2 - |\Delta|^2} \quad (3.61)$$

And a ground state energy:

$$-\frac{1}{2} \left(\sum_{\mathbf{k}\sigma} \xi_{(\sigma\mathbf{k})\sigma} + |\xi_{\mathbf{k}\uparrow} - \xi_{-\mathbf{k}\downarrow}| \right) \quad (3.62)$$

4 Kane-Mele Hubbard model

The Kane-Mele-Hubbard model is the Kane-Mele tight-binding model with an added Hubbard interaction:

$$H = \sum_{\langle i,j \rangle} c_{i\sigma}^\dagger c_{j\sigma} + i\lambda_{SO} \sum_{\langle\langle i,j \rangle\rangle} \nu_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow} \quad (4.1)$$

The slave-boson approach is a mean-field theory for this model obtained by decomposing the electron into spinon and bosonic holon and doublon degrees of freedom. The resulting spectrum was worked out in [2] and I'll follow that paper. The idea is that, for intermediate interaction strengths, the slave-boson treatment should be applicable since changing the occupancy of the electron sites is penalized and so becomes a virtual fluctuation.

We first need to decompose the electron operator $c_{i\sigma}^\dagger$ into spinon $f_{i\sigma}$, a holon h_i (a boson representing the unoccupied state $|\Omega\rangle$ at site i) and doublon d_i (a boson representing the doubly occupied state $|\uparrow\downarrow\rangle$ at i) degrees of freedom:

$$c_{i\sigma}^\dagger = f_{i\sigma}^\dagger h_i + \sigma d_i^\dagger f_{i,-\sigma} \quad (4.2)$$

This expression is just another way of describing what happens when an electron is added to the system, either:

1. a hole is eliminated and we get an extra spin, or
2. A doubly occupied state is created at the expense of a previously singly occupied state (which must have had the opposite spin to the electron just added).

Notice that only the singly occupied states $|\uparrow\rangle$ or $|\downarrow\rangle$ carry a non-zero spin - hence the spin degree of freedom has ‘seperated’ from the charge degree.

These operators satisfy canonical commutation/anti-commutation relations amongst themselves and commute with each other. These operators only interact with each other, then, through the constraints imposed by their interpretations. We have:

1. The completeness of the basis constraint:

$$h_i^\dagger h_i + \sum_{\sigma} f_{i\sigma}^\dagger f_{i\sigma} + d_i^\dagger d_i \stackrel{!}{=} 1 \quad (4.3)$$

2. The constraint arising from the the filling fraction of electrons when the material is doped away from half-filling by an amount x ⁹:

$$\sum_{\sigma} \langle c_{i\sigma}^\dagger c_{i\sigma} \rangle = f_{i\sigma}^\dagger f_{i\sigma} + 2d_i^\dagger d_i = 1 + \langle d_i^\dagger d_i \rangle - \langle h_i^\dagger h_i \rangle \equiv 1 + x \quad (4.4)$$

If we substitute this decomposition of the electron operator into the original Hamiltonian, we get expression (5) in the KMH paper [2]:

$$H = -t \sum_{\langle i,j \rangle} \left[\chi_{ij}^f \chi_{ji}^b + \Delta_{ij}^{f\dagger} \Delta_{ij}^b + h.c. \right] + \lambda_{SO} \sum_{\langle \langle i,j \rangle \rangle} \left[\tilde{\chi}_{ij}^f \tilde{\chi}_{ji}^b + \tilde{\Delta}_{ij}^{f\dagger} \tilde{\Delta}_{ij}^b + h.c. \right] + U \sum_i d_i^\dagger d_i \quad (4.5)$$

$$- \sum_i \lambda_i \left[h_i^\dagger h_i + \sum_{\sigma} f_{i\sigma}^\dagger f_{i\sigma} + d_i^\dagger d_i \right] - \sum_i \mu_i \left[d_i^\dagger d_i - h_i^\dagger h_i - x \right] \quad (4.6)$$

Where we have defined:

1. The fermion hopping $\chi_{ij}^f \equiv \sum_{\sigma} f_{i\sigma}^\dagger f_{i\sigma}$
2. The boson hopping $\chi_{ij}^b \equiv h_i^\dagger h_j - d_i^\dagger d_j$
3. The fermion pairing $\Delta_{ij}^f \equiv \sum_{\sigma} \sigma f_{i-\sigma} f_{j\sigma}$
4. The boson pairing $\Delta_{ij}^b \equiv d_i h_j + h_i d_j$
5. The fermion spin-orbit hopping $\tilde{\chi}_{ij}^b \equiv \sum_{\sigma} i \nu_{ij} f_{i\sigma}^\dagger f_{i\sigma}$
6. The boson spin-orbit hopping $\tilde{\chi}_{ij}^b \equiv h_i^\dagger h_j - d_i^\dagger d_j$

⁹Note that doublons only contain one of each spin

7. The fermion spin-orbit pairing $\tilde{\Delta}_{ij}^f \equiv \sum_{\sigma} i\nu_{ij}\sigma f_{i-\sigma}f_{j\sigma}$

8. The boson spin-orbit pairing $\tilde{\Delta}_{ij}^b \equiv d_i h_j + h_i d_j$

A mean-field approximation of this Hamiltonian around a stationary point of the path-integral in powers of the fields corresponding to these creation/annihilation operators gives an expression up to second order. In the form here, we have second order fermion or boson terms acting with their opposite boson or fermion terms respectively. Since the bosonic and fermionic operators commute any expectation value of a term pairing a fermionic or bosonic operator will factorise - $\langle f_{i\sigma}^{\dagger} d_j^{\dagger} \rangle = \langle f_{i\sigma}^{\dagger} \rangle \langle d_j^{\dagger} \rangle = 0$ since single fermion expectation operators must be zero to respect causality.¹⁰ This entails that, to first non-zero order in the fields, the spinon and boson degrees of freedom separate:

$$H = H_0 - t \sum_{\langle i,j \rangle} \left[\chi_{ij}^f \langle \chi_{ji}^b \rangle + \Delta_{ij}^{f\dagger} \langle \Delta_{ij}^b \rangle + h.c. \right] + \lambda_{SO} \sum_{\langle \langle i,j \rangle \rangle} \left[\tilde{\chi}_{ij}^f \langle \tilde{\chi}_{ji}^b \rangle + \tilde{\Delta}_{ij}^{f\dagger} \langle \tilde{\Delta}_{ij}^b \rangle + h.c. \right] \quad (4.7)$$

$$- t \sum_{\langle i,j \rangle} \left[\langle \chi_{ij}^f \rangle \chi_{ji}^b + \langle \Delta_{ij}^{f\dagger} \rangle \Delta_{ij}^b + h.c. \right] + \lambda_{SO} \sum_{\langle \langle i,j \rangle \rangle} \left[\langle \tilde{\chi}_{ij}^f \rangle \tilde{\chi}_{ji}^b + \langle \tilde{\Delta}_{ij}^{f\dagger} \rangle \tilde{\Delta}_{ij}^b + h.c. \right] \quad (4.8)$$

$$+ U \sum_i d_i^{\dagger} d_i - \sum_i \lambda_i \left[h_i^{\dagger} h_i + \sum_{\sigma} f_{i\sigma}^{\dagger} f_{i\sigma} + d_i^{\dagger} d_i \right] - \sum_i \mu_i \left[d_i^{\dagger} d_i - h_i^{\dagger} h_i - x \right] \quad (4.9)$$

With H_0 the expectation value of the Hamiltonian at the stationary point:

$$H_0 = -t \sum_{\langle i,j \rangle} \left[\langle \chi_{ij}^f \rangle \langle \chi_{ji}^b \rangle + \langle \Delta_{ij}^{f\dagger} \rangle \langle \Delta_{ij}^b \rangle + h.c. \right] \quad (4.10)$$

$$+ \lambda_{SO} \sum_{\langle \langle i,j \rangle \rangle} \left[\langle \tilde{\chi}_{ij}^f \rangle \langle \tilde{\chi}_{ji}^b \rangle + \langle \tilde{\Delta}_{ij}^{f\dagger} \rangle \langle \tilde{\Delta}_{ij}^b \rangle + h.c. \right] \quad (4.11)$$

$$+ U \sum_i \langle d_i^{\dagger} d_i \rangle - \sum_i \lambda_i \left[\langle h_i^{\dagger} h_i \rangle + \sum_{\sigma} \langle f_{i\sigma}^{\dagger} f_{i\sigma} \rangle + \langle d_i^{\dagger} d_i \rangle \right] \quad (4.12)$$

$$- \sum_i \mu_i \left[\langle d_i^{\dagger} d_i \rangle - \langle h_i^{\dagger} h_i \rangle - x \right] \quad (4.13)$$

We can then restrict ourselves to phases that preserve translational symmetry - which follows if we assume that all order parameters do not vary from site to site. For example:

$$\langle \chi_{ij}^b \rangle = \chi_b \quad (4.14)$$

for all sites i, j .¹¹

¹⁰I think!

¹¹Does having a point group symmetry allow you to make other choices here and would this lead to different

We can then take the Fourier transform of the mean field Hamiltonian $H = H_0 + H_b + H_f$ which has separated into a constant part H_0 , a part consisting only of bosonic operators H_b , and a part consisting only of fermionic operators H_f . The constant part is given by:

$$H_0 = -t \sum_{\langle ij \rangle} [\chi_b \chi_f + \Delta_f^\dagger \Delta_b + h.c.] + \lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} [\tilde{\chi}_b \tilde{\chi}_f + \tilde{\Delta}_f^\dagger \tilde{\Delta}_b + h.c.] \quad (4.15)$$

$$+ U n_d - \lambda \sum_i \left[n_h + \sum_\sigma n_{f\sigma} + n_d \right] - \mu \sum_i [n_d - n_h - x] \quad (4.16)$$

$$= 2z N t (\chi_b \chi_f + \Delta_f \Delta_b) + 2N(\lambda + x\mu) - 4z N \lambda_{SO} (\tilde{\chi}_b \tilde{\chi}_f + \tilde{\Delta}_f \tilde{\Delta}_b) \quad (4.17)$$

Where z is the coordination number (in the case of the Kane-Mele model the lattice is graphene and so $z = 3$).¹² The fermionic part is given by:

$$H_f = \sum_{\mathbf{k}\sigma\langle\alpha\rangle} \left[-t \chi_b g_{ij}(\mathbf{k}) f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{\mathbf{k}\sigma\alpha_j} - t \Delta_b g_{ij} \sigma(\mathbf{k}) f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{-\mathbf{k}(-\sigma)\alpha_j}^\dagger + h.c. \right] \quad (4.18)$$

$$+ \sum_{\mathbf{k}\sigma\langle\langle\alpha\rangle\rangle} \left[\lambda_{SO} \sigma \tilde{\chi}_b g_{1,ij}(\mathbf{k}) f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{\mathbf{k}\sigma\alpha_j} + \lambda_{SO} \tilde{\Delta}_b g_{1,ij} \nu_{ij}(\mathbf{k}) f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{\mathbf{k}(-\sigma)\alpha_j}^\dagger + h.c. \right] \quad (4.19)$$

$$- \lambda \sum_{\mathbf{k}\sigma\alpha} f_{\mathbf{k}\sigma\alpha}^\dagger f_{\mathbf{k}\sigma\alpha} \quad (4.20)$$

Where I have used $\langle \alpha \rangle$ to mean the non-spin indices (sublattice, valley etc...) on sites where the hopping takes place, and $\langle\langle \alpha \rangle\rangle$ for where spin-orbit hopping goes between. Note that the $g(\mathbf{k})$ is the tunnelling term between sites i and j - so $\mathbf{k}$ will be the reciprocal lattice vector between them. For a graphene base lattice this becomes equation (6) of the Kane-Mele Hubbard paper [2]:

$$H_f = \sum_{\mathbf{k}\sigma} \left[-t \chi_b g(\mathbf{k}) f_{\mathbf{k}\sigma A}^\dagger f_{\mathbf{k}\sigma B} - t \Delta_b g(\mathbf{k}) \sigma f_{\mathbf{k}\sigma A}^\dagger f_{\mathbf{k}(-\sigma)B}^\dagger + h.c. \right] \quad (4.21)$$

$$+ \sum_{\mathbf{k}\sigma} \lambda_{SO} \tilde{\chi}_b g_1(\mathbf{k}) \sigma \left[f_{\mathbf{k}\sigma A}^\dagger f_{\mathbf{k}\sigma A} - f_{\mathbf{k}\sigma B}^\dagger f_{\mathbf{k}\sigma B} \right] + \sum_{\mathbf{k}} \lambda_{SO} \tilde{\Delta}_b g_1(\mathbf{k}) \left[f_{\mathbf{k}\uparrow A}^\dagger f_{-\mathbf{k}\downarrow A}^\dagger - f_{\mathbf{k}\uparrow B}^\dagger f_{-\mathbf{k}\downarrow B}^\dagger \right] \quad (4.22)$$

$$- \lambda \sum_{\mathbf{k}\sigma\alpha} f_{\mathbf{k}\sigma\alpha}^\dagger f_{\mathbf{k}\sigma\alpha} + h.c. \quad (4.23)$$

An aside: You might wonder about the asymmetry of the hopping here - why do they have $f_{\mathbf{k}\uparrow A}^\dagger f_{-\mathbf{k}\downarrow A}^\dagger$ and not $f_{\mathbf{k}\downarrow A}^\dagger f_{-\mathbf{k}\uparrow A}^\dagger$? These terms end up being the same so long as $g_1(\mathbf{k}) = -g_1(-\mathbf{k})$ which is indeed the case for the Kane-Mele spin orbit hopping (but in general I think this is not true and you would have to include both!). So there is no asymmetry - it's just a more

phases?

¹²The factor of two in the hopping terms comes from the fact that the hopping is there and back, in the λ term it comes from the completeness of the basis and the average filling constraint.

convenient way of writing it.

Doing the same for the bosonic piece we get:

$$H_b = \sum_{\mathbf{k}\alpha} [U - \lambda - \mu - \lambda_{SO} \tilde{\chi}_f g_2(\mathbf{k})] d_{\mathbf{k}\alpha}^\dagger d_\alpha + [\mu - \lambda + \lambda_{SO} \tilde{\chi}_f g_2(\mathbf{k})] h_{\mathbf{k}\alpha}^\dagger h_{\mathbf{k}\alpha} \quad (4.24)$$

$$\sum_{\mathbf{k}} -t \chi_b g(\mathbf{k}) \left[h_{\mathbf{k}A}^\dagger h_{\mathbf{k}B} - d_{\mathbf{k}A}^\dagger d_{\mathbf{k}B} \right] + \sum_{\mathbf{k}} -t \Delta_f g(-\mathbf{k}) [d_{\mathbf{k}A} h_{-\mathbf{k}B} + h_{\mathbf{k}A} d_{-\mathbf{k}B}] \quad (4.25)$$

$$+ \sum_{\mathbf{k}\alpha} \lambda_{SO} \tilde{\Delta}_f g_2(\mathbf{k}) d_{\mathbf{k}\alpha} h_{-\mathbf{k}\alpha} \quad (4.26)$$

For reference, for graphene the hopping g 's are given by¹³

1. The hopping $g(\mathbf{k}) \equiv 1 + e^{-ik_2} + e^{i(k_1-k_2)}$ - Notice that $g(-\mathbf{k}) = g^*(\mathbf{k})$.
2. The spinon spin orbit hopping $g_1(\mathbf{k}) \equiv 2[\sin(k_2) - \sin(k_1) + \sin(k_2 - k_1)]$ - Notice that $g(-\mathbf{k}) = -g(\mathbf{k})$.
3. The bosonic spin orbit hopping¹⁴ $g_2(\mathbf{k}) \equiv 2[\cos(k_1) + \cos(k_2) + \cos(k_1 - k_2)]$ - Notice that $g_2(-\mathbf{k}) = g_2(\mathbf{k})$

We now need to solve H_0 . To implement the mean-field theory, we will need to do three things:

1. Solve the fermionic Hamiltonian, treating order parameters as unconstrained.
2. Solve the bosonic Hamiltonian, treating order parameters as unconstrained.
3. Solve the self-consistency equations to find the values of the order parameters.

4.1 Solving the Bosonic Hamiltonian

Now that we have obtained the Hamiltonian, we can proceed to solve it following the general procedure outlined in section 2. We're going to deal here with the case of **no spin orbit coupling** $\lambda_{SO} = 0$.

¹³Here k_i is a shorthand for $k_i a_i$ with a_i the vector between lattice sites.

¹⁴I think the difference comes from the fact that there is a ν_{ij} term in $\tilde{\chi}_f = \sum_{\sigma} i\nu_{ij} \sigma f_{i\sigma}^\dagger f_{j\sigma}$ but not in the bosonic counterpart $\tilde{\chi}_b = h_i d_j + d_i h_j$ and likewise for the pairing.

4.1.1 Writing down Nambu spinor

Following the convention we used before eq. (2.2), we can choose as a Nambu spinor:

$$\psi_{\mathbf{k}} = \begin{pmatrix} h_{\mathbf{k}A} \\ h_{\mathbf{k}B} \\ d_{-\mathbf{k}A} \\ d_{-\mathbf{k}B} \\ h_{\mathbf{k}A}^\dagger \\ h_{\mathbf{k}B}^\dagger \\ d_{-\mathbf{k}A}^\dagger \\ d_{-\mathbf{k}B}^\dagger \end{pmatrix} \quad (4.27)$$

4.1.2 Writing down the Hamiltonian Kernel

We now need to pick a Hamiltonian kernel $\mathcal{H}_b$ that will give us the terms in H_b :

$$H_b = \sum_{\mathbf{k}\alpha} (U - \lambda - \mu) d_{\mathbf{k}\alpha}^\dagger d_{\mathbf{k}\alpha} + (\mu - \lambda) h_{\mathbf{k}\alpha}^\dagger h_{\mathbf{k}\alpha} \quad (4.28)$$

$$\sum_{\mathbf{k}} -t\chi_b g(\mathbf{k}) \left[h_{\mathbf{k}A}^\dagger h_{\mathbf{k}B} - d_{\mathbf{k}A}^\dagger d_{\mathbf{k}B} \right] + \sum_{\mathbf{k}} -t\Delta_f g(-\mathbf{k}) [d_{\mathbf{k}A} h_{-\mathbf{k}B} + h_{\mathbf{k}A} d_{-\mathbf{k}B}] + h.c. \quad (4.29)$$

We can follow the same process as before, writing down a trial kernel for the diagonal, hopping and pairing terms and then checking that it reproduces the right terms. Since we know from the general procedure that the Hamiltonian kernel only has two independent block matrices ξ and Δ we can work with just these blocks. For the diagonal term, we can try:

$$\mathcal{H}_{b,D} = \begin{pmatrix} U - \lambda - \mu & 0 & 0 & 0 \\ 0 & U - \lambda - \mu & 0 & 0 \\ 0 & 0 & \mu - \lambda & 0 \\ 0 & 0 & 0 & \mu - \lambda \end{pmatrix} \quad (4.30)$$

If we multiply this out we get:

$$\sum_{\mathbf{k}} \begin{pmatrix} h_{\mathbf{k}A}^\dagger & h_{\mathbf{k}B}^\dagger & d_{-\mathbf{k}A}^\dagger & d_{-\mathbf{k}B}^\dagger \end{pmatrix} \begin{pmatrix} U - \lambda - \mu & 0 & 0 & 0 \\ 0 & U - \lambda - \mu & 0 & 0 \\ 0 & 0 & \mu - \lambda & 0 \\ 0 & 0 & 0 & \mu - \lambda \end{pmatrix} \begin{pmatrix} h_{\mathbf{k}A} \\ h_{\mathbf{k}B} \\ d_{-\mathbf{k}A} \\ d_{-\mathbf{k}B} \end{pmatrix} \quad (4.31)$$

$$= \sum_{\mathbf{k}} (U - \lambda - \mu)(h_{\mathbf{k}A}^\dagger h_{\mathbf{k}A} + h_{\mathbf{k}B}^\dagger h_{\mathbf{k}B}) + \sum_{-\mathbf{k}} (\mu - \lambda)(d_{\mathbf{k}A}^\dagger d_{\mathbf{k}A} + d_{\mathbf{k}B}^\dagger d_{\mathbf{k}B}) \quad (4.32)$$

Which is the diagonal term since $\mu - \lambda$ is $\mathbf{k}$ independent so we can re-index the last term. As before, we can put it in terms of Pauli matrices:

$$\boxed{\xi_D = D\nu^0\tau^0 + \Delta_0\nu^z\tau^0} \quad (4.33)$$

$$D \equiv \frac{D_d + D_h}{2} = \frac{(U - \lambda - \mu) + (\mu - \lambda)}{2} = \frac{U - 2\lambda}{2} \quad (4.34)$$

$$\Delta_0 \equiv \frac{D_d - D_h}{2} = \frac{(U - \lambda - \mu) - (\mu - \lambda)}{2} = \frac{U - 2\mu}{2} \quad (4.35)$$

Where the ν^i Pauli matrices act in the basis which specifies which block of two of h or d operators mix.

For the hopping term we can notice that:

$$\sum_{\mathbf{k}} -\chi_b(\mathbf{k})(h_{\mathbf{k}A}^\dagger h_{\mathbf{k}B} - d_{\mathbf{k}A}^\dagger d_{\mathbf{k}B}) + h.c. \quad (4.36)$$

is just:

$$\begin{pmatrix} h_{\mathbf{k}A}^\dagger & h_{\mathbf{k}B}^\dagger \end{pmatrix} \underbrace{\begin{pmatrix} 0 & -\chi_b^R - \chi_b^I \\ -\chi_b^R + \chi_b^I & 0 \end{pmatrix}}_{-\chi_b^R\sigma^x + \chi_b^I\tau^y} \begin{pmatrix} h_{\mathbf{k}A} \\ h_{\mathbf{k}B} \end{pmatrix} \quad (4.37)$$

and likewise for the term in the doublon operators. I have let

$$\chi_b(\mathbf{k}) \equiv t\chi_b g(\mathbf{k}) \quad (4.38)$$

$$\chi_b^R \equiv \text{Re}[\chi_b(\mathbf{k})] \quad (4.39)$$

$$\chi_b^I \equiv \text{Im}[\chi_b(\mathbf{k})] \quad (4.40)$$

to save space, TeX time, and sanity. This means we can see that if the original Hamiltonian

was given to us in terms of Pauli operators already (as it will be for SG135) we can just read off that the diagonal piece would be:

$$\nu^z(\text{real term in electron Pauli Hamiltonian}) + i\nu^0(\text{imaginary term}) \quad (4.41)$$

So in this case the hopping term is given by:

$$\boxed{\xi_D = -\chi_b^R \nu^z \tau^x + \chi_b^I \nu^0 \tau^y} \quad (4.42)$$

The pairing term is the trickiest. It's probably easiest to just write down an ansatz and see that it works (note that since the pairing block matrix should satisfy $\Delta = \zeta \Delta^T$ ($= \Delta^T$ for bosons) you only have to write down the top-left block here):

$$\mathcal{H}_{b,p} = \begin{pmatrix} 0 & 0 & 0 & -\Delta(\mathbf{k}) \\ 0 & 0 & -\Delta(-\mathbf{k}) & 0 \\ 0 & -\Delta(-\mathbf{k}) & 0 & 0 \\ -\Delta(\mathbf{k}) & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & -\Delta(\mathbf{k}) \\ 0 & 0 & -\Delta(\mathbf{k})^* & 0 \\ 0 & -\Delta(\mathbf{k})^* & 0 & 0 \\ -\Delta(\mathbf{k}) & 0 & 0 & 0 \end{pmatrix} \quad (4.43)$$

Using the same shorthand as before and the fact that $g(-\mathbf{k}) = g^*(\mathbf{k})$. In terms of Pauli matrices, this is:

$$\boxed{\Delta = -\Delta_b^R \nu^x \tau^x + i\Delta_b^I \nu^y \tau^y} \quad (4.44)$$

Putting all of these terms together, and remembering the general form of the bosonic BdG Hamiltonian from eq. (2.10):

$$\mathcal{H}_b = \begin{pmatrix} \xi & \Delta \\ \Delta^* & \xi^* \end{pmatrix} \quad (4.45)$$

We can write that the Hamiltonian kernel is given by:

$$\mathcal{H}_b = D\lambda^0 \nu^0 \tau^0 + \Delta_0 \lambda^0 \nu^z \tau^0 + [-\chi_b^R \lambda^0 \nu^z \tau^x + \chi_b^I \lambda^z \nu^0 \tau^y] + [-\Delta_b \lambda^x \nu^x \tau^x + i\Delta_b^I (i\lambda^y) \nu^x \tau^y] \quad (4.46)$$

Or

$$\boxed{\mathcal{H}_b = D\lambda^0 \nu^0 \tau^0 + \Delta_0 \lambda^0 \nu^z \tau^0 + [-\chi_b^R \lambda^0 \nu^z \tau^x + \chi_b^I \lambda^z \nu^0 \tau^y] + [-\Delta_b \lambda^x \nu^x \tau^x - \Delta_b^I \lambda^y \nu^y \tau^y]} \quad (4.47)$$

4.1.3 Diagonalizing the Hamiltonian Kernel

To diagonalize the bosonic Hamiltonian, we need to diagonalize $\lambda^z \mathcal{H}_b$:

$$\lambda^z \mathcal{H}_b = D\lambda^z \nu^0 \tau^0 + \Delta_0 \lambda^z \nu^z \tau^0 + [-\chi_b^R \lambda^z \nu^z \tau^x + \chi_b^I \lambda^0 \nu^0 \tau^y] + [-i\Delta_b \lambda^y \nu^x \tau^x + i\Delta_b^I \lambda^x \nu^y \tau^y] \quad (4.48)$$

Let's follow the same process as in the general procedure. First we draw out the network of anti-commutators - fig. 3. We can see that the splitting term $\Delta_0 \lambda^z \sigma^z \tau^0$ commutes with all the

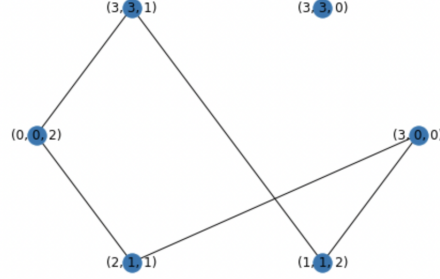


Figure 3: The network of anti-commutators of $\mathcal{H}_b$. Each node is a tensor product of Pauli matrices and an edge exists between any two nodes which anti-commute.

terms so we can separate it out. ince there are no remaining islands, that's as good as we can do. We can square the remaining kernel and keep track of any anti-commutators that turn out to be non-zero - fig. 4 We can then draw out the network of these commutators fig. 5. From

$$\begin{aligned} & ((3, 0, 0), (3, 3, 1)) \quad ((0, 3, 1), (2+0j)) \\ & ((3, 0, 0), (0, 0, 2)) \quad ((3, 0, 2), (2+0j)) \\ & ((3, 3, 1), (2, 1, 1)) \quad ((1, 2, 0), (2+0j)) \\ & ((0, 0, 2), (1, 2, 2)) \quad ((1, 2, 0), (2+0j)) \end{aligned}$$

Figure 4: The non-zero anti-commutators in the Hamiltonian kernel $\mathcal{H}_b$ when the Δ_0 term is removed. The first pair is the pair of the Pauli tensor indices which give a non-zero anti-commutator, the second pair are the Pauli tensor indices of the anti-commutator and the coefficient.

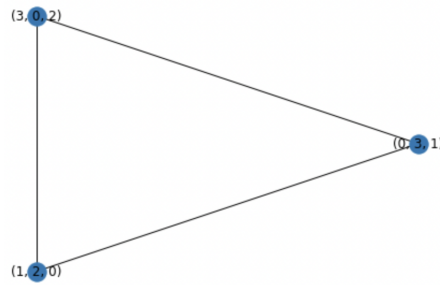


Figure 5: The network of anit-commutators of the anti-commutators of the terms in $\mathcal{H}'_b$. We can see that they form a single island so everything anti-commutes with everything. This means when we square one more time we'll get the identity, from which we can read off the eigenvalue spectrum of the Hamiltonian kernel.

here, we can read off the eigenvalue spectrum:

$$E = \pm_1 \Delta_0 \pm_2 \sqrt{D^2 + |\chi_b(\mathbf{k})|^2 - |\Delta_b(\mathbf{k})|^2} \pm_3 \sqrt{4(-D^2 \chi_b^R)^2 + 4(-D^2 \chi_b^I)^2 + 4(i\chi_b^R \Delta_b^R - i\chi_b^I \Delta_b^I)^2} \quad (4.49)$$

$$(4.50)$$

We can use trickery to simplify the last term:

$$(\chi^R \Delta^R - \chi^I \Delta^I)^2 = \chi^2 \Delta^2 (Re[g]^2 - Im[g]^2)^2 \quad (4.51)$$

$$= \chi^2 \Delta^2 (Re[g] + Im[g])^2 (Re[g] - Im[g])^2 = \chi^2 \Delta^2 (gg^*)^2 = |\chi|^2 |\Delta|^2 \quad (4.52)$$

Where we have used that the hopping and pairing have the same phase - i.e. we earlier defined:

$$\chi_b(\mathbf{k}) \equiv -t\chi_b g(\mathbf{k}) \quad (4.53)$$

$$\Delta_b(\mathbf{k}) \equiv -t\Delta_b g(\mathbf{k}) \quad (4.54)$$

So the eigenvalue spectrum simplifies down to:

$$E = \pm_1 \Delta_0 \pm_2 \sqrt{D^2 + |\chi_b(\mathbf{k})|^2 - |\Delta_b(\mathbf{k})|^2} \pm_3 2\sqrt{D^2 |\chi_b(\mathbf{k})| - |\chi|^2 |\Delta|^2} \quad (4.55)$$

$$= \pm_1 \Delta_0 \pm_2 \sqrt{\left(|\chi_b(\mathbf{k})| \pm_3 \sqrt{D^2 - |\Delta(\mathbf{k})|^2}\right)^2} \quad (4.56)$$

$$E = \boxed{\pm_1 \Delta_0 \pm_2 \left(|\chi_b(\mathbf{k})| \pm_3 \sqrt{D^2 - |\Delta(\mathbf{k})|^2}\right)} \quad (4.57)$$

We can check this in Mathematica -fig. 6.

```

In[ ]:= MatrixForm[H4]
Out[ ]:= MatrixForm[
  {
    {D1 + Δ0, -g χ, 0, 0, 0, 0, 0, -g Δ},
    {-χ Conjugate[g], D1 + Δ0, 0, 0, 0, 0, -Δ Conjugate[g], 0},
    {0, 0, D1 - Δ0, χ Conjugate[g], 0, 0, 0, 0},
    {0, 0, g χ, D1 - Δ0, -g Δ, 0, 0, 0},
    {0, 0, 0, Δ Conjugate[g], -D1 - Δ0, χ Conjugate[g], 0, 0},
    {0, 0, g Δ, 0, g χ, -D1 - Δ0, 0, 0},
    {Δ Conjugate[g], 0, 0, 0, 0, 0, -D1 + Δ0, -g χ},
    {0, 0, 0, 0, 0, 0, -χ Conjugate[g], -D1 + Δ0}
  }
]

In[ ]:= FullSimplify[Eigenvalues[H4, 1]]
Out[ ]:= {
  -√(g (-Δ² + χ²) Conjugate[g] + D1² - 2√(g χ² Conjugate[g] (-g Δ² Conjugate[g] + D1²)) - Δ0},
  ...
}

```

Figure 6: Mathematica diagonalization of the bosonic Hamiltonian.

4.1.4 Normal ordering the quasi-particle Hamiltonian

To normal order the Hamiltonian, we need to identify the positive part of the spectrum. As before there are two cases corresponding to a BEC or a BCS ground state¹⁵:

$$E = \begin{cases} \pm_1 \Delta_0 \pm_2 |\chi_b(\mathbf{k})| + \sqrt{D^2 - |\Delta(\mathbf{k})|^2}, & \text{for } \sqrt{D^2 - |\Delta(\mathbf{k})|^2} > |\chi_b(\mathbf{k})| \\ \pm_1 \Delta_0 + |\chi_b(\mathbf{k})| \pm_2 \sqrt{D^2 - |\Delta(\mathbf{k})|^2}, & \text{for } \sqrt{D^2 - |\Delta(\mathbf{k})|^2} < |\chi_b(\mathbf{k})| \end{cases} \quad (4.58)$$

Where the first has weak pairing relative to the hopping and so corresponds to the BEC case and the second to the BCS case since the pairing is strong. In the KMH paper [2], they choose the BEC case¹⁶. Following the general results, the normal ordered Hamiltonian is given by eq. (2.18)

$$H_b = \sum_n \epsilon_n b_n^\dagger b_n - \frac{1}{2} \left(\sum_n \epsilon_n + \text{Tr}\{\xi\} \right) \quad (4.59)$$

And so our ground state (in the BEC regime) is given by:

$$E_0 = -\frac{1}{2} \left(\sum_n \epsilon_n + \text{Tr}\{\xi\} \right) = -\frac{1}{2} \left(\sum_{\mathbf{k}} 2\sqrt{D^2 - |\Delta_b(\mathbf{k})|^2} + 2(U - 2\lambda) \right) \quad (4.60)$$

Putting in the original parameters of the Hamiltonian this recovers equation A4 of the KMH paper [2]:

$$E_0 = -U + 2\lambda - \sum_{\mathbf{k}} \sqrt{(U/2 - \lambda)^2 - |tg(\mathbf{k})\Delta_f|^2} \quad (4.61)$$

except for the 2λ (Not sure why this doesn't agree - looks like they might have put it back into the constant term).

4.2 Fermionic Hamiltonian

We can now go through the same process for the fermionic Hamiltonian.

¹⁵I'm not sure why you don't consider the Δ_0 term

¹⁶Sentence below equation A4.

4.2.1 Nambu spinor

We can make a similar choice of Nambu spinor to the bosonic case, but with the spin up/down now playing the role of doublon/holon

$$\psi_{\mathbf{k}} = \begin{pmatrix} f_{\mathbf{k}\uparrow A} \\ f_{\mathbf{k}\uparrow B} \\ f_{-\mathbf{k}\downarrow A} \\ f_{-\mathbf{k}\downarrow B} \\ f_{\mathbf{k}\uparrow A}^\dagger \\ f_{\mathbf{k}\uparrow B}^\dagger \\ f_{-\mathbf{k}\downarrow A}^\dagger \\ f_{-\mathbf{k}\downarrow B}^\dagger \end{pmatrix} \quad (4.62)$$

4.2.2 Writing down the Hamiltonian Kernel

We'll take it term by term, as before. The fermionic Hamiltonian with no spin orbit coupling is:

$$H_f = \sum_{\mathbf{k}\sigma} \left[-t\chi_b g(\mathbf{k}) f_{\mathbf{k}\sigma A}^\dagger f_{\mathbf{k}\sigma B} - t\Delta_b g(\mathbf{k}) \sigma f_{\mathbf{k}\sigma A}^\dagger f_{-\mathbf{k}(-\sigma)B}^\dagger + h.c. \right] - \lambda \sum_{\mathbf{k}\sigma\alpha} f_{\mathbf{k}\sigma\alpha}^\dagger f_{\mathbf{k}\sigma\alpha} \quad (4.63)$$

The diagonal term in the top-left kinetic block ξ is simply:

$$\begin{pmatrix} -\lambda & 0 & 0 & 0 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{pmatrix} = \boxed{-\lambda \sigma^0 \tau^0} \quad (4.64)$$

The hopping term in the kinetic block is:

$$\begin{pmatrix} 0 & -\chi & 0 & 0 \\ -\chi^* & 0 & 0 & 0 \\ 0 & 0 & 0 & -\chi^* \\ 0 & 0 & -\chi & 0 \end{pmatrix} = \boxed{-\chi_R \sigma^0 \tau^x + \chi_I \sigma^z \tau^y} \quad (4.65)$$

Where we will use analogous shorthands to the bosonic case:

$$\chi(\mathbf{k}) \equiv t\chi_b g(\mathbf{k}) \quad (4.66)$$

$$\Delta(\mathbf{k}) \equiv t\Delta_b g(\mathbf{k}) \quad (4.67)$$

For the pairing term, we can again use the fact for the fermionic case the pairing block Δ satisfies $\Delta = -\Delta^T$. Since we now pick up a sign when moving operators through each other, we also get a sign change. Notice also that the σ in the pairing term in the Hamiltonian means that the two blocks should have inverse signs to each other. We can verify that the term below will work:

$$\begin{pmatrix} 0 & 0 & 0 & -\Delta \\ 0 & 0 & -\Delta^* & 0 \\ 0 & \Delta^* & 0 & 0 \\ \Delta & 0 & 0 & 0 \end{pmatrix} = \boxed{-i\Delta_R \sigma^y \tau^x + \Delta_I \sigma^x \tau^y} \quad (4.68)$$

For fermions, the general procedure tells us that the blocks of the matrix are related by eq. (2.10):

$$\mathcal{H}_f = \begin{pmatrix} \xi & \Delta \\ -\Delta^* & -\xi^* \end{pmatrix} \quad (4.69)$$

And so we can put together our three terms to get:

$$\mathcal{H}_f = -\lambda(\lambda^z \sigma^0 \tau^0) - \chi_R \lambda^z \sigma^0 \tau^x + \chi_I \lambda^0 \sigma^z \tau^y - i\Delta_R (i\lambda^y) \sigma^y \tau^x + \Delta_I \lambda^x \sigma^x \tau^y \quad (4.70)$$

$$\boxed{\mathcal{H}_f = -\lambda(\lambda^z \sigma^0 \tau^0) + (-\chi_R \lambda^z \sigma^0 \tau^x + \chi_I \lambda^0 \sigma^z \tau^y) + (\Delta_R \lambda^y \sigma^y \tau^x + \Delta_I \lambda^x \sigma^x \tau^y)} \quad (4.71)$$

4.2.3 Diagonalizing the Hamiltonian Kernel

Now we can diagonalize, following the same procedure as before. Mapping out the network of anti-commutators in fig. 7, we see that there are no islands¹⁷ As before we can calculate all of

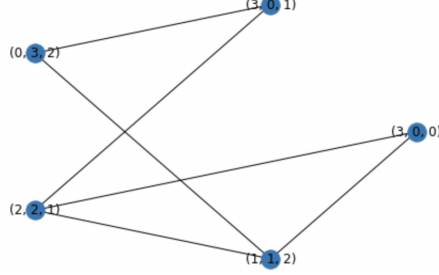


Figure 7: The network of anti-commutators of $\mathcal{H}_f$. Each node is a tensor product of Pauli matrices and an edge exists between any two nodes which anti-commute. We can see that there are no islands so we can't separate off any of the terms.

the anti-commutators - fig. 8. From the non-zero anti-commutators we can infer that:

$$\begin{aligned} &((3, 0, 0), (3, 0, 1)) ((0, 0, 1), (2+0j)) \\ &((3, 0, 0), (0, 3, 2)) ((3, 3, 2), (2+0j)) \\ &((3, 0, 1), (1, 1, 2)) ((2, 1, 3), (-2+0j)) \\ &((0, 3, 2), (2, 2, 1)) ((2, 1, 3), (-2+0j)) \end{aligned}$$

Figure 8: The non-zero anti-commutators of $\mathcal{H}_f$.

$$\mathcal{H}_f^2 = (\lambda^2 + \chi_R^2 + \chi_I^2 + \Delta_R^2 + \Delta_I^2) \mathbb{1} \quad (4.72)$$

$$+ 2(-\lambda)(-\chi_R)\lambda^0\sigma^0\tau^x + 2(-\lambda)(\chi_I)\lambda^z\sigma^z\tau^y - 2[(-\chi_R)(\Delta_I) + \chi_I\Delta_I] \lambda^x\sigma^x\tau^z \quad (4.73)$$

$$= [\lambda^2 + |\chi|^2 + |\Delta|^2] \mathbb{1} + 2\lambda\chi_R(\lambda^0\sigma^0\tau^x) - 2\lambda\chi^I(\lambda^z\sigma^z\tau^y) - 2[-\chi_R\Delta_I + \chi_I\Delta_R] \quad (4.74)$$

We can use a similar trick to what we did for bosons to show that the last term is zero¹⁸:

$$\chi_R\Delta_I\chi_I\Delta_R = \chi'\Delta'(-\text{Re}[g]\text{Im}[g] + \text{Im}[g]\text{Re}[g]) = 0 \quad (4.75)$$

Where I've again used the fact that the phase of χ and Δ is the same, which follows from their original definitions:

$$\chi = \underbrace{t\chi_f}_{\chi'} g(\mathbf{k}) \quad (4.76)$$

$$\Delta = \underbrace{t\Delta_f}_{\Delta'} g(\mathbf{k}) \quad (4.77)$$

¹⁷Which is what we'd expect since there is no splitting term in this case.

¹⁸It's interesting that for fermions the pairing and the hopping don't couple to each other - why is that?

Taking out the identity terms since it commutes with all the others, the remaining matrices anti-commute so $\mathcal{H}_f^4$ is proportional to the identity -fig. 9. And so we have the spectrum:

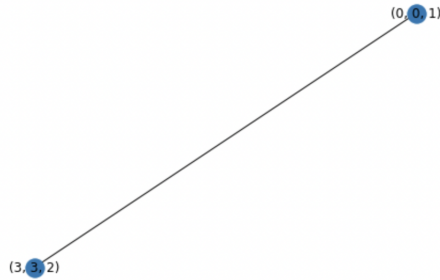


Figure 9: The non-zero anti-commutators of $\mathcal{H}_f^2$. Everything anti-commutes and so the square is proportional to the identity. Note that if the last term wasn't zero, then we would still have all three matrices anti-commute like in the bosonic case.

$$E = \pm_1 \sqrt{(\lambda^2 + |\chi|^2 + |\Delta|^2) \pm_2 2\sqrt{\lambda^2 |\chi|^2}} \quad (4.78)$$

$$= \pm_1 \sqrt{(\lambda^2 + |\chi|^2 + |\Delta|^2) \pm_2 2\lambda |\chi|} \quad (4.79)$$

$$(4.80)$$

And so the eigenvalue spectrum is given by:

$$\boxed{E = \pm_1 \sqrt{(\lambda^2 + |\chi|^2 + |\Delta|^2) \pm_2 2\lambda |\chi|}} \quad (4.81)$$

Notice that the fermionic term has twice the degeneracy of the bosonic case. This makes intuitive sense - there is no spin-splitting in the fermions, whereas the interaction term U splits the holon and doublon energies.

4.2.4 Normal ordering the Hamiltonian

As before, the doubling of the Fock space means only the positive energies correspond to excitation energies. Since there is an overall minus sign there is only one choice for the positive excitation energies:

$$\epsilon_{\pm}(\mathbf{k}) = \sqrt{(\lambda^2 + |\chi|^2 + |\Delta|^2) \pm_2 2\lambda |\chi|} \quad (4.82)$$

And so given the general form of the fermionic normal-ordered Hamiltonian eq. (2.18):

$$H = \sum_n \epsilon_n b_n^\dagger b_n - \frac{1}{2} \left(\sum_n \epsilon_n - \text{Tr}\{\xi\} \right) \quad (4.83)$$

We can write down:

$$H = \sum_{\mathbf{k}\pm} \epsilon_{\pm}(\mathbf{k}) b_{\mathbf{k}\pm}^{\dagger} b_{\mathbf{k}\pm} - \frac{1}{2} \left(\sum_{\mathbf{k}} \sqrt{\lambda^2 + |\chi|^2 + |\Delta|^2 - 2\lambda|\chi|} + \sqrt{\lambda^2 + |\chi|^2 + |\Delta|^2 + 2\lambda|\chi|} + 4\lambda \right) \quad (4.84)$$

And so, in the original parameters of the Hamiltonian, the ground state is given by:

$$E_0 = -2\lambda - \frac{1}{2} \sum_{\mathbf{k}} \sqrt{\lambda^2 + |t\chi_b g(\mathbf{k})|^2 + |t\Delta_b g(\mathbf{k})|^2 + 2\lambda|t\chi_b g(\mathbf{k})|} + \sqrt{\lambda^2 + |t\chi_b g(\mathbf{k})|^2 + |t\Delta_b|^2 - 2\lambda|t\chi_b g(\mathbf{k})|} \quad (4.85)$$

Which agrees with A5 in the KMH paper [2] up to the -2λ . The 2λ will cancel with the $+2\lambda$ in the bosonic spectrum - eq. (4.61) (I guess that's why it's not in the expression in the paper).

4.3 Constant energy

The final term in the ground state energy is that which comes from the zero-th order term in the order parameter operators. The Hamiltonian was given in eq. (4.15):

$$H_0 = -t \sum_{\langle ij \rangle} [\chi_b \chi_f + \Delta_f^{\dagger} \Delta_b + h.c.] + \lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} [\tilde{\chi}_b \tilde{\chi}_f + \tilde{\Delta}_f^{\dagger} \tilde{\Delta}_b + h.c.] \quad (4.86)$$

$$+ U n_d - \lambda \sum_i \left[n_h + \sum_{\sigma} n_{f\sigma} + n_d \right] - \mu \sum_i [n_d - n_h - x] \quad (4.87)$$

$$= 2zNt(\chi_b \chi_f + \Delta_f \Delta_b) + 2N(\lambda + x\mu) - 4zN\lambda_{SO}(\tilde{\chi}_b \tilde{\chi}_f + \tilde{\Delta}_f \tilde{\Delta}_b) \quad (4.88)$$

Setting the spin-orbit coupling equal to zero we have:

$$H_0 = 2zNt(\chi_b \chi_f + \Delta_f \Delta_b) + 2N(\lambda + x\mu) \quad (4.89)$$

Which implies it contributes an energy:

$$E_c = 2zNt(\chi_b \chi_f + \Delta_f \Delta_b) + 2N(\lambda + x\mu) \quad (4.90)$$

to the ground state. If we want to take into account the possibility of forming a BEC, we will need to introduce order parameters for the expectation of $d_{\mathbf{k}=0,A}$, $d_{\mathbf{k}=0,B}$, $h_{\mathbf{k}=0,A}$, $h_{\mathbf{k}=0,B}$ for the number of bosonic operators in the $\mathbf{k} = 0$ state where the BEC must form for a translationally invariant system. Notice that these are real c-numbers so the hermitian conjugation doesn't change them.

If we split these terms of in the bosonic Hamiltonian we will also have a contribution:

$$H_b = (U - \lambda - \mu)d_{\mathbf{k}=0\alpha}^\dagger d_{\mathbf{k}=0\alpha} + (\mu - \lambda)h_{\mathbf{k}=0\alpha}^\dagger h_{\mathbf{k}=0\alpha} \quad (4.91)$$

$$- t\chi_b \underbrace{g(\mathbf{k}=0)}_{=3} \left[h_{\mathbf{k}=0A}^\dagger h_{\mathbf{k}=0B} - d_{\mathbf{k}=0A}^\dagger d_{\mathbf{k}=0B} \right] - t\Delta_f g(\mathbf{k}=0) [d_{\mathbf{k}=0A} h_{-\mathbf{k}=0B} + h_{\mathbf{k}=0A} d_{-\mathbf{k}=0B}] + h.c. \quad (4.92)$$

$$= (U - \lambda - \mu)(d_A^2 + d_B^2) + (\mu - \lambda)(h_A^2 + h_B^2) - 2(3)\chi_b(h_A h_B - d_A d_B) - 2(3)\Delta_f(d_A h_B + d_B h_A) \quad (4.93)$$

Where the factor of 2 comes from counting the hermitian conjugate and the factor of three comes from the fact that $g(\mathbf{k}) \equiv e^{ik_1} + e^{-ik_2} + e^{ik_2-k_1}$ and so $g(\mathbf{k}=0) = 3$.

4.4 Numerical plots

To check that we have it right and see the full band structure, we can do the diagonalization step numerically. We get:

4.5 Self-consistency equations

We can now collect together the ground state contribution from the constant term, the fermionic Hamiltonian and the bosonic Hamiltonian to get:

$$E_0 = 6t(\chi_b \chi_f + \Delta_f \Delta_b) + 2N(\lambda + x\mu) \quad (4.94)$$

$$+ (U - \lambda - \mu)(d_A^2 + d_B^2) + (\mu - \lambda)(h_A^2 + h_B^2) - 6\chi_b(h_A h_B - d_A d_B) - 6\Delta_f(d_A h_B + d_B h_A) \quad (4.95)$$

$$- \frac{1}{2} \sum_{\mathbf{k}} \sqrt{\lambda^2 + |t\chi_b g(\mathbf{k})|^2 + |t\Delta_b g(\mathbf{k})|^2 + 2\lambda|t\chi_b g(\mathbf{k})|} + \sqrt{\lambda^2 + |t\chi_b g(\mathbf{k})|^2 + |t\Delta_b g(\mathbf{k})|^2 - 2\lambda|t\chi_b g(\mathbf{k})|} \quad (4.96)$$

$$- \sum_{\mathbf{k}} \sqrt{(U/2 - \lambda)^2 - |tg(\mathbf{k})\Delta_f|^2} \quad (4.97)$$

To find the value of the order parameters in terms of the free parameters of the Hamiltonian, MFT requires us to take minimize this energy with respect to each of the Lagrange multipliers λ, μ and order parameters. Setting the partial derivatives equal to zero gives the constraint equations.

5 SG135

We can follow the same procedure to come up with a slave-boson theory¹⁹ for a more complicated crystal symmetry. We can pretty much copy over the formalism of the KMH paper [2], except now we have more complicated hoppings.

In [3], the authors find a Double-Dirac semi-metal for crystals with space group 135 and 130 symmetries. Let's take the case of SG135 first. With no spin-orbit coupling, the Hamiltonian kernel is:

$$\mathcal{H} = t_{xy}\tau^x \cos(k_x/2) \cos(k_y/2) + t_z\mu^x \cos(k_z/2) \quad (5.1)$$

$$+ t'_1\mu^z(\cos(k_x) - \cos(k_y)) + t'_2\tau^x\tau^y \sin(k_x/2) \sin(k_y/2) \cos(k_z/2) \quad (5.2)$$

Where μ^i, τ^i are Pauli matrices in the basis of two sublattices: It's easy to visualise the hoppings.

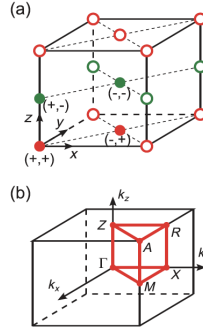


Figure 10: The two sub-lattices that define μ and τ . μ corresponds to the sublattice translated by half a lattice vector up from the parent lattice, and the tau sublattice corresponds to the sub-lattice translated in x and y by half a lattice vector. The bottom figure is the associated Brillouin zone.

The last term, say, corresponds to hoppings backwards and forwards that interchange both the μ and τ sub-lattices. To write down the slave-boson theory, we can decompose the electron operator in the same way before and so we have the same expression:

$$H = \sum_{\langle i,j \rangle} t_{ij} \left[\chi_{ij}^f \chi_{ji}^b + \Delta_{ij}^{f\dagger} \Delta_{ij}^b + h.c. \right] + \sum_{\langle\langle i,j \rangle\rangle} \lambda_{SO,ij} \left[\tilde{\chi}_{ij}^f \tilde{\chi}_{ji}^b + \tilde{\Delta}_{ij}^{f\dagger} \tilde{\Delta}_{ij}^b + h.c. \right] + U \sum_i d_i^\dagger d_i \quad (5.3)$$

$$- \sum_i \lambda_i \left[h_i^\dagger h_i + \sum_\sigma f_{i\sigma}^\dagger f_{i\sigma} + d_i^\dagger d_i \right] - \sum_i \mu_i \left[d_i^\dagger d_i - h_i^\dagger h_i - x \right] \quad (5.4)$$

except now the angled brackets correspond to all of these hoppings. We can hence proceed as before - solving the bosonic and fermionic Hamiltonians individually following the procedure

¹⁹There's still a question about whether the slave boson or slave rotor approach works better here

laid out in the general section [2](#).

5.1 Bosonic Hamiltonian

The bosonic Hamiltonian is:

$$H_b = \sum_{\mathbf{k}\alpha} [U - \lambda - \mu] d_{\mathbf{k}\alpha}^\dagger d_{\mathbf{k}\alpha} + [\mu - \lambda] h_{\mathbf{k}\alpha}^\dagger h_{\mathbf{k}\alpha} \quad (5.5)$$

$$\sum_{\mathbf{k},i,j} -t_{ij} \chi_b g_{ij}(\mathbf{k}) \left[h_{\mathbf{k}\alpha_i}^\dagger h_{\mathbf{k}\alpha_j} - d_{\mathbf{k}\alpha_i}^\dagger d_{\mathbf{k}\alpha_j} \right] + \sum_{\mathbf{k},i,j} -t_{ij} \Delta_f g_{ij}(-\mathbf{k}) \left[d_{\mathbf{k}\alpha_i} h_{-\mathbf{k}\alpha_j} + h_{\mathbf{k}\alpha_i} d_{-\mathbf{k}\alpha_j} \right] \quad (5.6)$$

5.1.1 Nambu spinor

Following the convention we have been using, the Nambu spinor is:

$$\psi_{\mathbf{k}} = \begin{pmatrix} h_{\mathbf{k}A,A} \\ h_{\mathbf{k}A,B} \\ h_{\mathbf{k}B,A} \\ h_{\mathbf{k}B,B} \\ d_{-\mathbf{k}A,A} \\ d_{-\mathbf{k}A,B} \\ d_{-\mathbf{k}B,A} \\ d_{-\mathbf{k}B,B} \\ h.c. \end{pmatrix} \quad (5.7)$$

Where the next eight rows are just the Hermitian conjugate of the first eight.

5.1.2 Hamiltonian kernel

We now proceed as before, writing down the diagonal, hopping and pairing terms in term. The diagonal term for the kinetic energy block ξ is:

$$D\nu^0 \mu^0 \tau^0 + \Delta_0 \nu^z \mu^0 \tau^0 \quad (5.8)$$

Where as before:

$$D \equiv \frac{D_d + D_h}{2} = \frac{U - 2\lambda}{2} \quad (5.9)$$

$$\Delta \equiv \frac{D_d - D_h}{2} = \frac{U - 2\mu}{2} \quad (5.10)$$

We can then write down the hopping and pairing terms for each of the four terms in the original Hamiltonian. There are two helpful properties of the Hamiltonian that makes this a little easier:

1. All terms in the Hamiltonian are inversion symmetric $\mathcal{H}(\mathbf{k}) = \mathcal{H}(-\mathbf{k})$
2. All terms in the Hamiltonian are real.

This means that the hopping term for the top-left kinetic block ξ is just given by:

$$\nu^z(\text{term in the starting Hamiltonian kernel}) \quad (5.11)$$

And the pairing term for the top-right pairing block Δ is given by:

$$\nu^z(\text{term in the starting Hamiltonian kernel}) \quad (5.12)$$

Along with the general form of the bosonic Hamiltonian kernel from the general procedure:

$$\begin{pmatrix} \xi & \Delta \\ \Delta^* & \xi^* \end{pmatrix} \quad (5.13)$$

This means our terms in the full Hamiltonian kernel are given by:

$$\text{Hopping} = \lambda^0 \nu^z (\text{Term in original Hamiltonian}) \quad (5.14)$$

$$\text{Pairing} = \lambda^x \nu^x (\text{Term in original Hamiltonian}) \quad (5.15)$$

This simplification has happened because the two complications we had before arose from: choosing half of the terms to be indexed by $-\mathbf{k}$, from terms being in the wrong order, and from conjugation. Neither are an issue here because of the inversion symmetry, the fact that all off-diagonal bosonic terms commute, and because all the terms are real. ²⁰ This allows us to write down the Hamiltonian kernel immediately:

²⁰The fact that they are real is a special property of the cubic lattice. If you need to convince yourself just draw out the hoppings and you'll see that each hopping from a site to another has a hopping in the opposite direction. This is not the case for graphene, and so writing down the terms was more difficult for the Kane-Mele Hubbard model.

$$\mathcal{H}_b(\mathbf{k}) = D\lambda^0\nu^0\mu^0\tau^0 + \Delta_0\lambda^0\nu^z\mu^0\tau^0 \quad (5.16)$$

$$+ (\chi_{xy}\lambda^0\nu^z + \Delta_{xy}\lambda^x\nu^x)\mu^0\tau^x + (\chi_z\lambda^0\nu^z + \Delta_z\lambda^x\nu^x)\mu^x\tau^0 \quad (5.17)$$

$$+ (\chi_1\lambda^0\nu^z + \Delta_{xy}\lambda^x\nu^x)\mu^z\tau^0 + (\chi_2\lambda^0\nu^z + \Delta_2\lambda^x\nu^x)\mu^y\tau^y \quad (5.18)$$

Where I've used a similar shorthand to before - absorbing the hopping term $g(\mathbf{k})$ and the hopping parameter t into the order parameters:

$$\chi_{ij} \equiv t_{ij}\chi_f g_{ij}(\mathbf{k}) \quad (5.19)$$

$$\Delta_{ij} \equiv t_{ij}\Delta_f g_{ij}(\mathbf{k}) \quad (5.20)$$

5.1.3 Diagonalizing the Hamiltonian kernel

To diagonalize the bosonic Hamiltonian by a canonical transformation, we need to diagonalize instead:

$$\lambda^z \mathcal{H}_b(\mathbf{k}) = D\lambda^z\nu^0\mu^0\tau^0 + \Delta\lambda^z\nu^z\mu^0\tau^0 \quad (5.21)$$

$$+ (\chi_{xy}\lambda^z\nu^z + i\Delta_{xy}\lambda^y\nu^x)\mu^0\tau^x + (\chi_z\lambda^z\nu^z + i\Delta_z\lambda^y\nu^x)\mu^x\tau^0 \quad (5.22)$$

$$+ (\chi_1\lambda^z\nu^z + i\Delta_{xy}\lambda^y\nu^x)\mu^z\tau^0 + (\chi_2\lambda^z\nu^z + i\Delta_2\lambda^y\nu^x)\mu^y\tau^y \quad (5.23)$$

The usual procedure of squaring the Hamiltonian until you get something proportional to the identity doesn't work here, and so the diagonalization ends up being a mess. Interestingly, squaring the Hamiltonian will work if we drop any of the terms that are not the $\mu^y\tau^y$ term.

5.1.4 Diagonalizing the linearized Hamiltonian

If we linearize the Hamiltonian around the $A = (\pi, \pi, \pi)$ point of the Brillouin zones, we remove two of the terms and the resulting Hamiltonian diagonalizes straightforwardly. Expanding the tunnelling terms around this point:

$$g_{xy} = \cos(k_x/2) \cos(k_y/2)|_{(\pi, \pi, \pi)} = \mathcal{O}(k^2) \quad (5.24)$$

$$g_z = \cos(k_z/2)|_{(\pi, \pi, \pi)} = -\frac{1}{4}k_z + \mathcal{O}(k^2) \quad (5.25)$$

$$g_1 = (\cos(k_x) - \cos(k_y))|_{(\pi, \pi, \pi)} = \mathcal{O}(k^2) \quad (5.26)$$

$$g_2 = \sin(k_x/2) \sin(k_y/2) \cos(k_z/2)|_{(\pi, \pi, \pi)} = -\frac{1}{4}k_z + \mathcal{O}(k^2) \quad (5.27)$$

And so, to linear order around the A point, the slave-boson Hamiltonian kernel is:

$$\lambda^z \mathcal{H}_b(\mathbf{k}) = D\lambda^z \nu^0 \mu^0 \tau^0 + \Delta_0 \lambda^z \nu^z \mu^0 \tau^0 \quad (5.28)$$

$$+ (\chi_z \lambda^z \nu^z + i\Delta_z \lambda^y \nu^x) \mu^x \tau^0 + (\chi_2 \lambda^z \nu^z + i\Delta_2 \lambda^y \nu^x) \mu^y \tau^y \quad (5.29)$$

To solve this we can follow the same procedure as before. The splitting commutes with everything - fig. 11. The non-zero anti-commutators are given in fig. 12. And we can in turn draw the

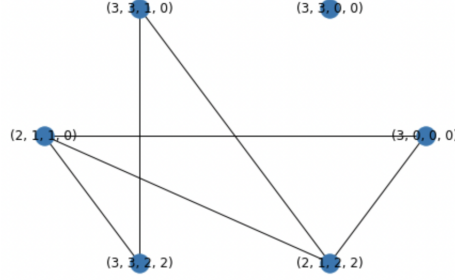


Figure 11: The network of anti-commutators of $\mathcal{H}$. As before only the splitting commutes with everything

$$\begin{aligned} &((3, 0, 0, 0), (3, 3, 1, 0)) ((0, 3, 1, 0), (2+0j)) \\ &((3, 0, 0, 0), (3, 3, 2, 2)) ((0, 3, 2, 2), (2+0j)) \\ &((3, 3, 1, 0), (2, 1, 1, 0)) ((1, 2, 0, 0), (2+0j)) \\ &((3, 3, 2, 2), (2, 1, 2, 2)) ((1, 2, 0, 0), (2+0j)) \end{aligned}$$

Figure 12: The non-zero anti-commutators of the Hamiltonian kernel linearized about the A -point.

network of anti-commutators of the non-zero anti-commutators fig. 13. We see that all of them anti-commute and so $\mathcal{H}_b^2$ is proportional to the identity. We can hence read off the spectrum:

$$\pm_1 \Delta_0 \pm_2 \sqrt{D^2 + \chi_z^2 + \chi_2^2 - \Delta_z^2 - \Delta_2^2 \pm_3 2\sqrt{\chi_z^2(D^2 - \Delta_z^2) + \chi_2^2(D^2 - \Delta_2^2) - 2\chi_2\chi_z\Delta_z\Delta_2}} \quad (5.30)$$

We can verify that the diagonalization at least is right by brute force diagonalization in Mathe-

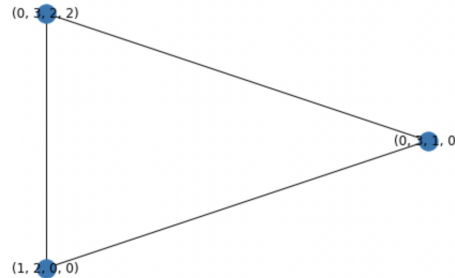


Figure 13: The network of anti-commutators for the square of the Hamiltonian kernel linearized around A .

matica - fig. 14. Remember that one of these $\pm$'s goes away because only the positive eigenvalues correspond to excitation energies of the quasi-particles.

```

HDDSMBA = Refine[D KroneckerProduct[PauliMatrix[3], PauliMatrix[0], PauliMatrix[0], PauliMatrix[0]] +
  Δ0 KroneckerProduct[PauliMatrix[3], PauliMatrix[3], PauliMatrix[0], PauliMatrix[0]] + Xz KroneckerProduct[PauliMatrix[3], PauliMatrix[3],
  Δz KroneckerProduct[PauliMatrix[2], PauliMatrix[1], PauliMatrix[1], PauliMatrix[0]] +
  X2 KroneckerProduct[PauliMatrix[3], PauliMatrix[3], PauliMatrix[2], PauliMatrix[2]] +
  Δ2 KroneckerProduct[PauliMatrix[2], PauliMatrix[1], PauliMatrix[2], PauliMatrix[2]],
  {Xz Element Reals, Δz Element Reals, X2 Element Reals, Δ2 Element Reals, D Element Reals, Δ0 Element Reals}]

In[42]:= MatrixForm[FullSimplify[HDDSMBA]]

```

$$\begin{pmatrix}
D + \Delta_0 & 0 & X_z & -X_z & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Delta_0 & -\Delta_2 \\
0 & D + \Delta_0 & X_z & X_z & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Delta_2 & \Delta_0 \\
X_z & X_z & D + \Delta_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Delta_0 & 0 & 0 \\
-X_z & X_z & 0 & D + \Delta_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\Delta_2 & \Delta_0 & 0 & 0 \\
0 & 0 & 0 & 0 & D - \Delta_0 & 0 & -X_z & X_z & 0 & 0 & \Delta_0 & -\Delta_2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & D - \Delta_0 & -X_z & -X_z & 0 & 0 & \Delta_2 & \Delta_0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -X_z & -X_z & D - \Delta_0 & 0 & \Delta_0 & \Delta_2 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & X_z & -X_z & 0 & D - \Delta_0 & -\Delta_2 & \Delta_0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\Delta_0 & \Delta_2 & -D - \Delta_0 & 0 & -X_z & X_z & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\Delta_2 & -\Delta_0 & 0 & -D - \Delta_0 & -X_z & -X_z & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\Delta_0 & -\Delta_2 & 0 & 0 & -X_z & -X_z & -D - \Delta_0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \Delta_2 & -\Delta_0 & 0 & 0 & X_z & -X_z & -D - \Delta_0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\Delta_0 & \Delta_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -D + \Delta_0 & 0 & X_z & -X_z & 0 \\
0 & 0 & -\Delta_2 & -\Delta_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -D + \Delta_0 & X_z & X_z & 0 \\
-\Delta_0 & -\Delta_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & X_z & X_z & -D + \Delta_0 & 0 \\
\Delta_2 & -\Delta_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -X_z & X_z & 0 & -D + \Delta_0 & 0
\end{pmatrix}$$

```

In[46]:= FullSimplify[Eigenvalues[HDDSMBA]]
Out[46]= {
  -Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 - 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 - √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  -Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2),
  Δ0 + √(D2 - Δ22 - Δz2 + X22 + Xz2 + 2 √(D2 - Δ22) X22 - 2 Δ2 Δz X2 Xz + (D2 - Δz2) Xz2 }

```

Figure 14: Brute force diagonalization around the A point.

5.1.5 Normal ordering

We can write down the ground state using the general result:

$$E_0 = -\frac{1}{2} \left(\sum_{vebk} 4(U - 2\lambda) + \sum_{n,\mathbf{k}} \epsilon_n \right) = -2UN + 4\lambda N - \sum_{\mathbf{k}n} \frac{\epsilon_n(\mathbf{k})}{2} \quad (5.31)$$

5.1.6 Numerical diagonalization

We can verify our results and see the full band structure by numerically diagonalizing the Hamiltonian. Let's first set the diagonal term D , the holon/doublon splitting Δ_0 and all the pairing order parameters Δ_i to 0 in order to compare with the band structure in the Dirac semi-

metal paper [3]. For SG135, they get fig. 15 and for the same choice of t parameters (remember that they also have spin orbit coupling terms) I have fig. 16.

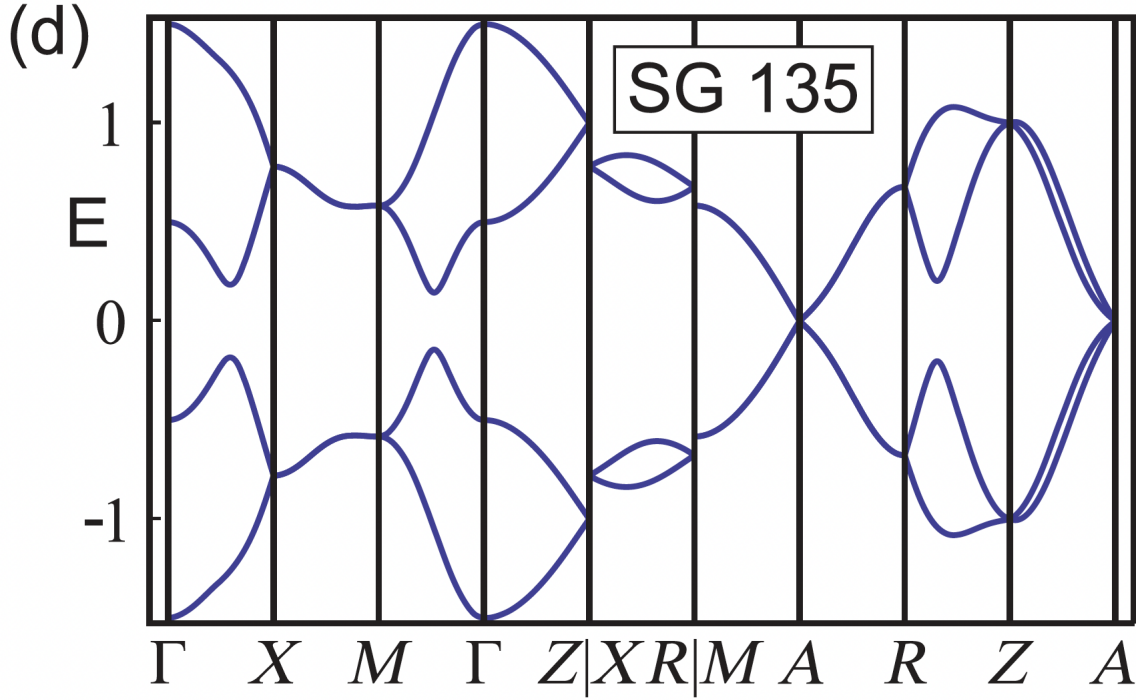


Figure 15: Numerical band structure for full bosonic Hamiltonian with parameter values $t_x y = 1, t_z = 0.5, t_1 = 0.3, t_2 = 0.3, \chi_i = 1, \Delta_i = 0, D = 0$. Along all the high symmetry paths except the last one from Z to A the degeneracy is half of what it is for the non-interacting case.

If we turn on the pairing terms, we get fig. 17.

5.2 Fermionic Hamiltonian

We can go through the same process for the fermionic Hamiltonian. It's just slightly more complicated because of the sign change on commuting.

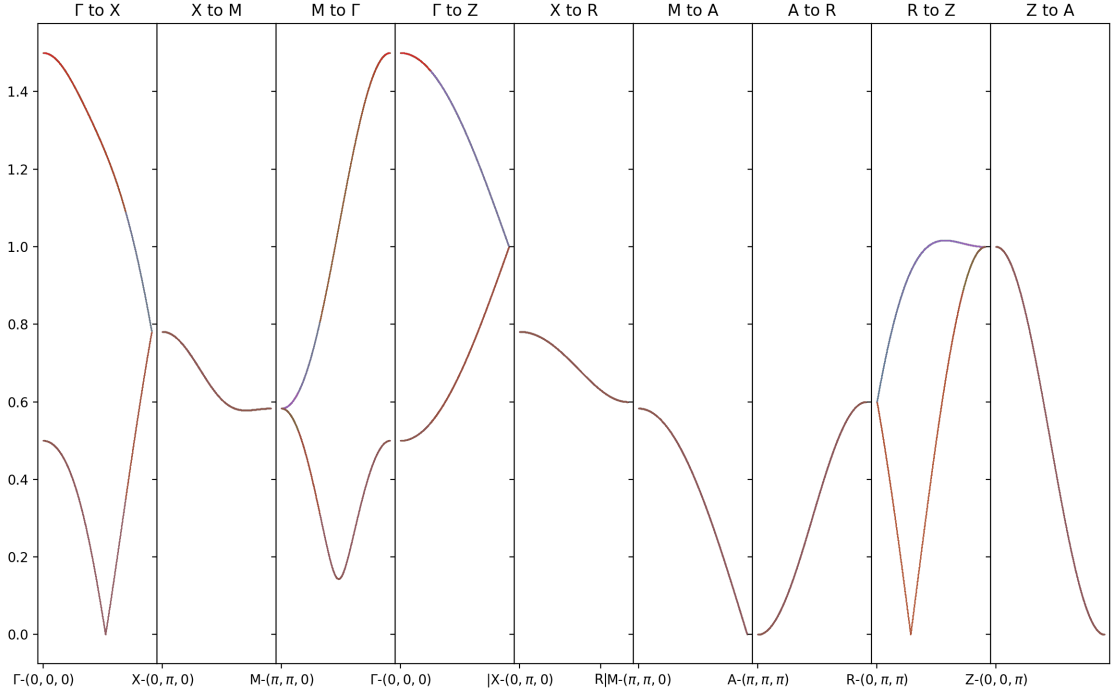


Figure 16: Numerical band structure for full bosonic Hamiltonian with parameter values $t_x y = 1, t_z = 0.5, t_1 = 0.3, t_2 = 0.3, \chi_i = 1, \Delta_i = 0, D = 0$. Along all the high symmetry paths except the last one from Z to A the degeneracy is half of what it is for the non-interacting case.

5.2.1 Nambu spinor

We can choose a Nambu spinor:

$$\psi_{\mathbf{k}} = \begin{pmatrix} f_{\mathbf{k}\uparrow,A,A} \\ f_{\mathbf{k}\uparrow,A,B} \\ f_{\mathbf{k}\uparrow,B,A} \\ f_{\mathbf{k}\uparrow,B,B} \\ f_{-\mathbf{k}\downarrow,A,A} \\ f_{-\mathbf{k}\downarrow,A,B} \\ f_{-\mathbf{k}\downarrow,B,A} \\ f_{-\mathbf{k}\downarrow,B,B} \\ h.c. \end{pmatrix} \quad (5.32)$$

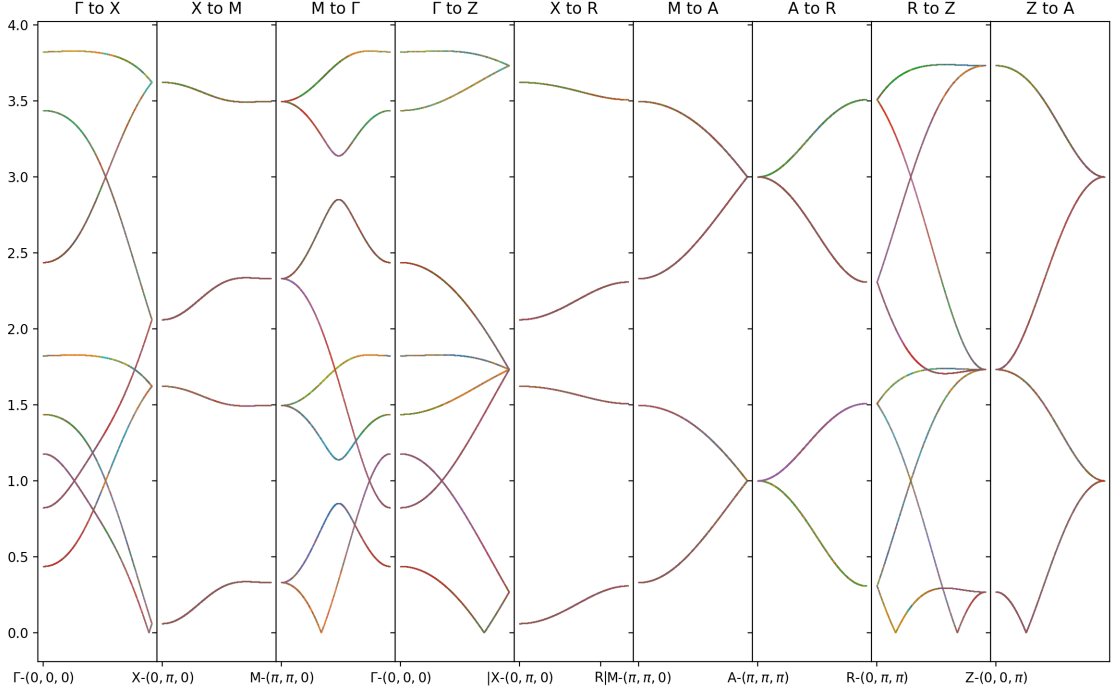


Figure 17: Numerical band structure for full bosonic Hamiltonian with parameter values $t_x y = 1, t_z = 0.5, t_1 = 0.3, t_2 = 0.3, \chi_i = 1, \Delta_i = 1, D = 2, \Delta_0 = 1$.

Where the Hermitian conjugate indicates that the rows below are the hermitian conjugate of those above.

5.2.2 Writing down the Hamiltonian kernel

We start from the fermionic part of the mean-field theory:

$$H_f = \sum_{\mathbf{k}\sigma ij} \left[-t_{ij} \chi_{b,ij} g_{ij}(\mathbf{k}) f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{\mathbf{k}\sigma\alpha_j} - t \Delta_{b,ij} g_{ij}(\mathbf{k}) \sigma f_{\mathbf{k}\sigma\alpha_i}^\dagger f_{-\mathbf{k}(-\sigma)\alpha_j}^\dagger + h.c. \right] - \lambda \sum_{\mathbf{k}\sigma\alpha} f_{\mathbf{k}\sigma\alpha}^\dagger f_{\mathbf{k}\sigma\alpha} \quad (5.33)$$

We can straightforwardly write down the diagonal term of the top-left kinetic block ξ as before:

$$\xi_{f,d} = -\lambda \sigma^0 \mu^0 \tau^0 \quad (5.34)$$

The hopping terms for the upper-left kinetic block χ are:

$$\sigma^0 (\text{Hamiltonian kernel term}) \quad (5.35)$$

The sign change in the fermion commutator is cancelled by the factor of σ in the pairing term of the Hamiltonian so the upper-right pairing block is given by²¹:

$$i\sigma^y(\text{Hopping term in original Hamiltonian kernel}) \quad (5.36)$$

Using the structure of the Hamiltonian kernel for fermions:

$$\begin{pmatrix} \xi & \Delta \\ -\Delta^* & -\xi^* \end{pmatrix} \quad (5.37)$$

The terms are given by²²:

$$\text{Hopping} = \lambda^z \sigma^0 (\text{Hopping term in original Hamiltonian kernel}) \quad (5.38)$$

$$\text{Pairing} = -\lambda^y \sigma^y (\text{Hopping term in original Hamiltonian kernel}) \quad (5.39)$$

And so we can write down the fermionic Hamiltonian kernel:

$$\mathcal{H}_f(\mathbf{k}) = -\lambda(\lambda^z \sigma^0 \mu^0 \tau^0) \quad (5.40)$$

$$+ (\lambda^z \sigma^0 \chi_{xy} - \lambda^y \sigma^y \Delta_{xy}) \mu^0 \tau^x + (\lambda^z \sigma^0 \chi_z - \lambda^y \sigma^y \Delta_z) \mu^x \tau^0 \quad (5.41)$$

$$+ (\lambda^z \sigma^0 \chi_1 - \lambda^y \sigma^y \Delta_1) \mu^z \tau^0 + (\lambda^z \sigma^0 \chi_2 - \lambda^y \sigma^y \Delta_2) \mu^y \tau^y \quad (5.42)$$

Using the same shorthand as in the bosonic case.

5.2.3 Diagonalizing the Hamiltonian kernel linearized about the A point

As before, trying to diagonalize this by squaring the Hamiltonian repeatedly will not lead to the identity. Diagonalizing by brute force gives a nasty expression -fig. 18. If we linearize around the A point, then it will diagonalize straightforwardly. We already expanded the tunnelling terms around the A -point for the bosonic part:

$$g_{xy} = \cos(k_x/2) \cos(k_y/2)|_{(\pi,\pi,\pi)} = \mathcal{O}(k^2) \quad (5.43)$$

$$g_z = \cos(k_z/2)|_{(\pi,\pi,\pi)} = -\frac{1}{4}k_z + \mathcal{O}(k^2) \quad (5.44)$$

$$g_1 = (\cos(k_x) - \cos(k_y))|_{(\pi,\pi,\pi)} = \mathcal{O}(k^2) \quad (5.45)$$

$$g_2 = \sin(k_x/2) \sin(k_y/2) \cos(k_z/2)|_{(\pi,\pi,\pi)} = -\frac{1}{4}k_z + \mathcal{O}(k^2) \quad (5.46)$$

²¹Equivalently, this is what you would get if you imposed the condition that $\Delta = \zeta \Delta^T$, which makes sense since this condition accounts for the fermion property. If the original spin-hopping term was itself spin dependent then it would be more complicated.

²²Note Two factors of i in the pairing - one from $i\sigma^y$ one from $i\lambda^y$

```
In[2]:= MatrixForm[FullSimplify[HDDSMf]]
```

ixForm=

$-\lambda + X_1$	X_{xy}	X_z	$-X_2$	0	0	0	0	0	0	0	0	Δ_1	Δ_z	Δ_z	$-\Delta_2$
X_{xy}	$-\lambda + X_1$	X_2	X_z	0	0	0	0	0	0	0	0	Δ_2	Δ_1	Δ_z	Δ_z
X_z	X_2	$-\lambda - X_1$	X_{xy}	0	0	0	0	0	0	0	0	Δ_z	Δ_2	$-\Delta_1$	Δ_z
$-X_2$	X_z	X_{xy}	$-\lambda - X_1$	0	0	0	0	0	0	0	0	$-\Delta_2$	Δ_z	Δ_z	$-\Delta_1$
0	0	0	0	$-\lambda + X_1$	X_{xy}	X_z	$-X_2$	$-\Delta_1$	$-\Delta_z$	$-\Delta_z$	Δ_2	0	0	0	0
0	0	0	0	X_{xy}	$-\lambda + X_1$	X_2	X_z	$-\Delta_z$	$-\Delta_1$	$-\Delta_2$	$-\Delta_2$	0	0	0	0
0	0	0	0	0	X_z	X_2	$-\lambda - X_1$	X_{xy}	$-\Delta_z$	$-\Delta_2$	Δ_1	$-\Delta_2$	0	0	0
0	0	0	0	0	$-X_2$	X_z	X_{xy}	$-\lambda - X_1$	Δ_2	$-\Delta_z$	Δ_1	0	0	0	0
0	0	0	0	0	$-\Delta_1$	$-\Delta_z$	$-\Delta_2$	Δ_2	$\lambda - X_1$	$-X_{xy}$	$-X_z$	X_2	0	0	0
0	0	0	0	0	$-\Delta_z$	$-\Delta_1$	$-\Delta_2$	$-\Delta_z$	$-X_{xy}$	$\lambda - X_1$	$-X_2$	$-X_z$	0	0	0
0	0	0	0	0	$-\Delta_z$	$-\Delta_2$	Δ_1	$-\Delta_z$	$-X_z$	$\lambda - X_1$	$-X_{xy}$	0	0	0	0
0	0	0	0	0	Δ_2	$-\Delta_z$	$-\Delta_z$	Δ_1	X_2	$-X_z$	$-X_{xy}$	$\lambda + X_1$	0	0	0
Δ_1	Δ_z	Δ_z	$-\Delta_2$	0	0	0	0	0	0	0	0	$\lambda - X_1$	$-X_{xy}$	$-X_z$	X_2
Δ_z	Δ_1	Δ_2	Δ_z	0	0	0	0	0	0	0	0	$-X_{xy}$	$\lambda - X_1$	$-X_2$	$-X_z$
Δ_z	Δ_z	$-\Delta_1$	Δ_2	0	0	0	0	0	0	0	0	$-X_z$	$-X_2$	$\lambda + X_1$	$-X_{xy}$
$-\Delta_2$	Δ_z	Δ_z	Δ_z	$-\Delta_1$	0	0	0	0	0	0	0	X_2	$-X_z$	$-X_{xy}$	$\lambda + X_1$

[illegible]

And so the fermionic Hamiltonian kernel linearized around the A point becomes:

$$+ (\lambda^z \sigma^0 \chi_z - \lambda^y \sigma^y \Delta_z) \mu^x \tau^0 + (\lambda^z \sigma^0 \chi_2 - \lambda^y \sigma^y \Delta_2) \mu^y \tau^y \quad (5.48)$$

$$E = \pm_1 \sqrt{\lambda^2 + \Delta_2^2 + \chi_2^2 + \Delta_z^2 + \chi_z^2} \pm_2 2\sqrt{\lambda^2(\chi_2^2 + \chi_z^2) + \Delta_z^2\chi_2^2 + \Delta_2^2\chi_z^2 - 2\Delta_2\Delta_z\chi_2\chi_z} \quad (5.49)$$
$$\epsilon_{\pm} = \sqrt{\lambda^2 + \Delta_2^2 + \chi_2^2 + \Delta_z^2 + \chi_z^2 \pm 2\sqrt{\lambda^2(\chi_2^2 + \chi_z^2) + \Delta_z^2\chi_2^2 + \Delta_2^2\chi_z^2 - 2\Delta_2\Delta_z\chi_2\chi_z}} \quad (5.50)$$

Which is four-fold degenerate. The ground state is then:

$$E_0 = -\frac{1}{2} \left(\sum_n \epsilon_n - \text{Tr}\{\xi\} \right) \quad (5.51)$$

$$= -\frac{1}{2} \left(\sum_{\mathbf{k}} 4(\epsilon_+ + \epsilon_-) + 8\lambda \right) = -4\lambda N - \sum_{\mathbf{k}} 2(\epsilon_+(\mathbf{k}) + \epsilon_-(\mathbf{k})) \quad (5.52)$$

5.2.4 Numerical diagonalization

Like for the bosonic case, we can verify our result numerically and see the full band structure. If we turn off the diagonal term λ and set the pairings $\Delta_i = 0$ we have fig. 19. If we turn on

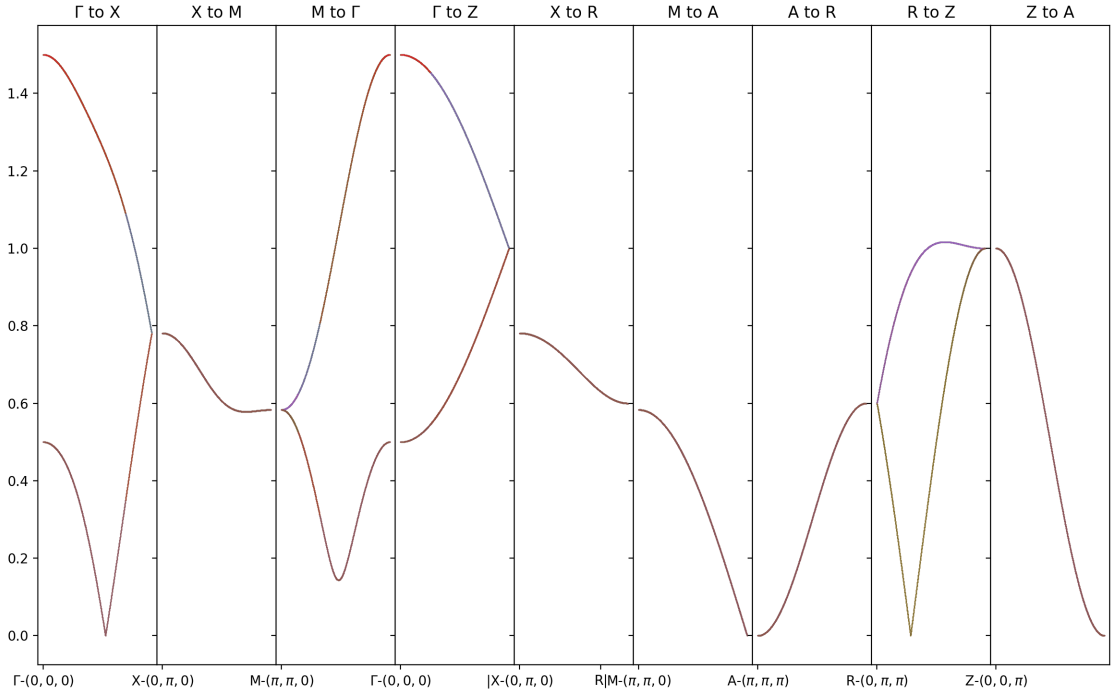


Figure 19: Numerical band structure for the fermionic Hamiltonian with no pairing. Parameter values $t_{xy} = 1, t_z = 0.5, t_1 = 0.3, t_2 = 0.3, \chi_i = 1, \lambda = 0, \Delta_i = 0$. It's the same as the fermionic Hamiltonian because they only differ in the pairing and the possibility of forming a BEC.

the pairings Δ_i and the diagonal term λ we get fig. 20. This agrees with our linearized result around the A-point.

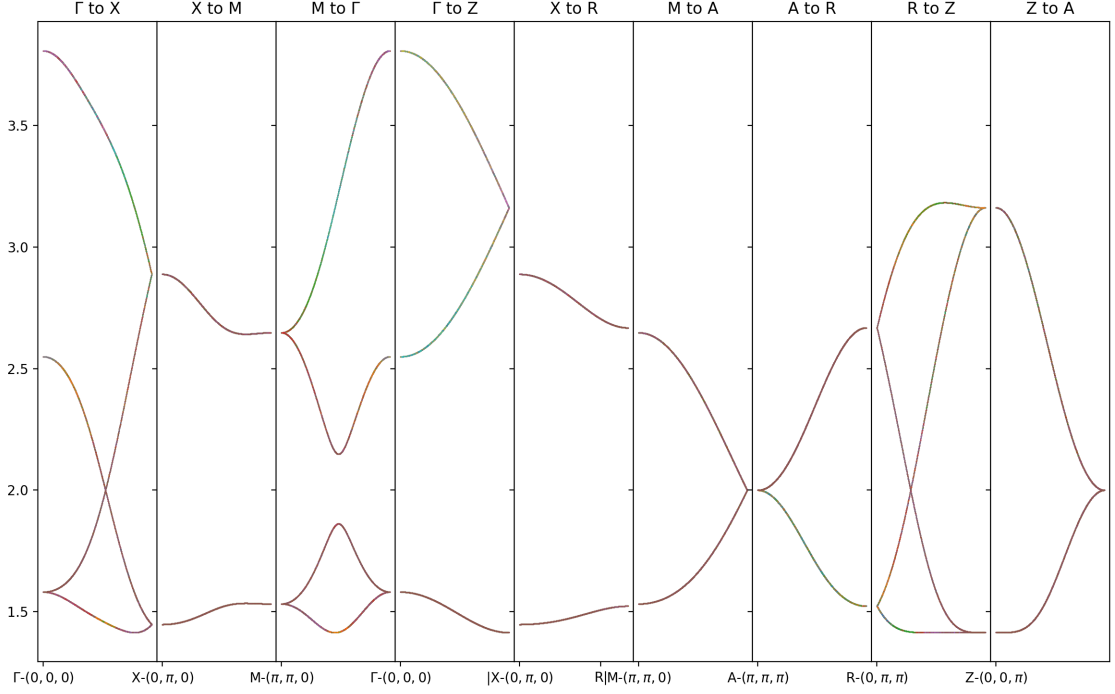


Figure 20: Numerical band structure for the fermionic Hamiltonian with no pairing. Parameter values $t_{xy} = 1, t_z = 0.5, t_1 = 0.3, t_2 = 0.3, \chi_i = 1, \lambda = 2, \Delta_i = 1$. The degeneracy is half of what it is in the DDSM paper fig. 15 except along the final high symmetry path from Z to A .

5.3 Self-consistency equation

To get to the self-consistency equation, we need to write down the constant energy term. This is just as in the Kane-Mele model, except now there are more order parameters and the coordination number is different:

$$E_c = \sum_i 2z_i N t (\chi_{b,i} \chi_{f,i} + \Delta_{f,i} \Delta_{b,i}) + 4N(\lambda + x\mu) \quad (5.53)$$

$$+ \sum_i (U - \lambda - \mu)(d_{\alpha_i}^2 + d_{\beta_i}^2) + (\mu - \lambda)(h_{\alpha_i}^2 + h_{\beta_i}^2) \quad (5.54)$$

$$- \sum_i 2z_i \chi_{f,i} (h_{\alpha_i} h_{\beta_i} - d_{\alpha_i} d_{\beta_i}) - 2z_i \Delta_{f,i} (d_{\alpha_i} h_{\beta_i} + d_{\beta_i} h_{\alpha_i}) \quad (5.55)$$

Where α_i, β_i denote the spin and sublattice indices for each hopping indexed by i , and z_i is the coordination number of each hopping (here all equal to 4).

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